

EXISTENCE OF MINIMAL MODELS FOR VARIETIES OF LOG GENERAL TYPE

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ABSTRACT. We prove that the canonical ring of a smooth projective variety is finitely generated.

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1. INTRODUCTION

The purpose of this paper is to prove the following result in birational algebraic geometry:

Theorem 1.1. *Let (X, Δ) be a projective kawamata log terminal pair. If Δ is big and $K_X + \Delta$ is pseudo-effective then $K_X + \Delta$ has a log terminal model.*

In particular, it follows that if $K_X + \Delta$ is big then it has a log canonical model and the canonical ring is finitely generated. So, for example, if X is a smooth projective variety, then the ring

$$R(X, K_X) = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(mK_X)),$$

is finitely generated.

The birational classification of complex projective surfaces was understood by the Italian Algebraic Geometers in the early 20th century: If X is a smooth complex projective surface of positive Kodaira dimension, that is $\kappa(X, K_X) \geq 0$, then there is a unique smooth surface Y birational to X such that the canonical class K_Y is nef (that is $K_Y \cdot C \geq 0$ for any curve $C \subset Y$). Y is obtained from X simply by contracting all -1 -curves, that is all smooth rational curves E with $K_X \cdot E = -1$. If, on the other hand, $\kappa(X, K_X) = -\infty$, then X is birational to either $\mathbb{P}^2$ or a ruled surface over a curve of genus $g > 0$.

The Minimal Model Program aims to generalise the classification of complex projective surfaces to higher dimensional varieties. The main goal of this program is to show that given any n -dimensional complex projective variety X , we have:

- If $\kappa(X, K_X) \geq 0$, then there exists a **minimal model**, that is a variety Y birational to X such that K_Y is nef.
- If $\kappa(X, K_X) = -\infty$, then there is a variety Y birational to X which admits a **Fano fibration**, that is a morphism $Y \rightarrow Z$ whose fibres F have ample anticanonical class $-K_F$.

It is straightforward to exhibit 3-folds which have no smooth minimal model, and so one must allow varieties X with singularities. However, these singularities cannot be arbitrary. At the very minimum, we must

still be able to compute $K_X \cdot C$ for any curve $C \subset X$. So, we insist that K_X is $\mathbb{Q}$ -Cartier (or sometimes we require the stronger property that X is $\mathbb{Q}$ -factorial). We also require that X and Y have the same pluricanonical forms. This condition is essentially equivalent to requiring that the induced birational map $\phi: X \dashrightarrow Y$ is K_X -negative.

There are two natural ways to construct the minimal model (it turns out that if one can always construct a minimal model, then we can always construct a Fano fibration). Since one of the main ideas of this paper is to blend the techniques of both methods, we describe both methods.

The first method is to use the ideas behind finite generation. If the canonical ring

$$R(X, K_X) = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(mK_X)),$$

is finitely generated then the canonical model Y is nothing more than the Proj of R . It is then automatic that the induced rational map $\phi: X \dashrightarrow Y$ is K_X -negative.

The other natural way to ensure that ϕ is K_X -negative is to factor ϕ into a sequence of elementary steps all of which are K_X -negative. We now explain one way to achieve this factorisation:

If K_X is not nef, then, by the Cone Theorem, there is a rational curve $C \subset X$ such that $K_X \cdot C < 0$ and a morphism $f: X \rightarrow Z$ which is surjective, with connected fibres, on to a normal projective variety and which contracts an irreducible curve D if and only if $[D] \in \mathbb{R}^+[C] \subset N_1(X)$. Note that $\rho(X/Z) = 1$ and $-K_X$ is f -ample. We have the following possibilities:

- If $\dim Z < \dim X$, this is the required Fano fibration.
- If $\dim Z = \dim X$ and f contracts a divisor, then we say that f is a divisorial contraction and we replace X by Z .
- If $\dim Z = \dim X$ and f does not contract a divisor, then we say that f is a small contraction. In this case K_Z is not $\mathbb{Q}$ -Cartier, so that we can not replace X by Z . Instead, we would like to replace $f: X \rightarrow Z$ by its flip $f^+: X^+ \rightarrow Z$ where X^+ is isomorphic to X in codimension 1 and K_{X^+} is f^+ -ample. In other words, we wish to replace some K_X -negative curves by K_{X^+} -positive curves.

The idea is to simply repeat the above procedure until we obtain either a minimal model or a Fano fibration. For this procedure to succeed, we must show that flips always exist and that they eventually terminate. Since the Picard number $\rho(X)$ drops by one after each divisorial

contraction, there can be at most finitely many divisorial contractions. So we must show that there is no infinite sequence of flips.

This program was successfully completed for 3-folds in the 1980's by the work of Kawamata, Kollár, Mori, Reid, Shokurov and others. In particular, the existence of 3-fold flips was proved by Mori in [30].

Naturally, one would hope to extend these results to dimension 4 and higher proceeding by induction on the dimension.

Recently, Shokurov has shown the existence of flips in dimension 4 [37] and Hacon and McKernan [13] have shown that assuming the minimal model program in dimension $n - 1$ (or even better simply finiteness of minimal models in dimension $n - 1$), then flips exist in dimension n . Thus we get an inductive approach to finite generation.

Unfortunately the problem of showing termination of an arbitrary sequence of flips seems to be a very difficult problem and in dimension ≥ 4 only some partial answers are available. Kawamata, Matsuda and Matsuki proved [22] the termination of terminal 4-fold flips, Matsuki has shown [29] the termination of terminal 4-fold flops and Fujino has shown [8] the termination of canonical 4-fold (log) flips. Alexeev, Hacon and Kawamata [2] have shown the termination of kawamata log terminal 4-fold flips when $-(K_X + \Delta)$ is big and the existence of minimal models of kawamata log terminal 4-folds when $K_X + \Delta$ is big by showing the termination of a certain sequence of flips (those that appear in the MMP with scaling). However, it is known that termination of flips follows from two natural conjectures on the behaviour of the log discrepancies of n -dimensional pairs (namely the ascending chain condition for minimal log discrepancies and semicontinuity of log discrepancies, cf. [38]). Moreover, if $\kappa(X, K_X + \Delta) \geq 0$, Birkar has shown [4] that it suffices to establish acc for log canonical thresholds and the MMP in dimension one less.

We now turn to the main result of the paper:

Theorem 1.2. *Let (X, Δ) be a kawamata log terminal pair, where $K_X + \Delta$ is $\mathbb{R}$ -Cartier. Let $\pi: X \rightarrow U$ be a proper morphism, where U is an algebraic space.*

If either Δ is π -big and $K_X + \Delta$ is π -pseudo-effective or $K_X + \Delta$ is π -big, then

- (1) $K_X + \Delta$ has a log terminal model over U ,
- (2) $K_X + \Delta$ has a log canonical model over U , and
- (3) if $K_X + \Delta$ is $\mathbb{Q}$ -Cartier, then the $\mathcal{O}_U$ -algebra

$$\bigoplus_{m \in \mathbb{N}} \pi_* \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor),$$

is finitely generated.

We now present some consequences of (1.2), most of which are well known. Even though we do not prove termination of flips, we are able to derive many of the consequences of the existence of the MMP. In many cases we do not state the strongest results possible; anyone interested in further applications is directed to the references. We group these consequences under different headings.

1.1. Minimal models. An immediate consequence of (1.2) is:

Corollary 1.1.1. *Let X be a smooth projective variety of general type. Then*

- (1) *X has a minimal model,*
- (2) *X has a canonical model,*
- (3) *the ring*

$$\bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(mK_X)),$$

is finitely generated, and

- (4) *X has a model with a Kähler-Einstein metric.*

Note that (4) follows from (2) and Theorem D of [7].

Corollary 1.1.2. *Let (X, Δ) be a kawamata log terminal pair, where $K_X + \Delta$ is $\mathbb{Q}$ -Cartier.*

Then the ring

$$R(X, K_X + \Delta) = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor)),$$

is finitely generated.

We will now turn our attention to the geography of minimal models. It is well known that log terminal models are not unique. The first natural question about log terminal models is to understand how any two are related. In fact there is a very simple connection:

Corollary 1.1.3. *Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Suppose that $K_X + \Delta$ is kawamata log terminal and Δ is big over U . Let $\phi_i: X \dashrightarrow Y_i$, $i = 1$ and 2 be two log terminal models of (X, Δ) over U . Let $\Gamma_i = \phi_{i*}\Delta$.*

Then the birational map $Y_1 \dashrightarrow Y_2$ is the composition of a sequence of $(K_{Y_1} + \Gamma_1)$ -flips over U .

The next natural problem is to understand how many different models there are. Even if log terminal models are not unique, there are only finitely many. In fact Shokurov realised that much more is true,

that the dependence on Δ is well-behaved. To explain this, we need some definitions:

Definition 1.1.4. *Let $\pi: X \longrightarrow U$ be a projective morphism of normal quasi-projective varieties, and let V be a finite dimensional affine subspace of the real vector space of Weil divisors on X which is defined over the rationals. Define*

$$\begin{aligned}\mathcal{L} &= \{ \Delta \in V \mid K_X + \Delta \text{ is log canonical} \}, \\ \mathcal{N} &= \{ \Delta \in \mathcal{L} \mid K_X + \Delta \text{ is nef} \}.\end{aligned}$$

Moreover, fixing an $\mathbb{R}$ -divisor A , define

$$\begin{aligned}V_A &= \{ \Delta \mid \Delta = A + B, B \in V \}, \\ \mathcal{L}_A &= \{ \Delta \in V_A \mid K_X + \Delta \text{ is log canonical} \}, \\ \mathcal{P}_A &= \{ \Delta \in \mathcal{L}_A \mid K_X + \Delta \text{ is pseudo-effective} \}, \\ \mathcal{N}_A &= \{ \Delta \in \mathcal{L}_A \mid K_X + \Delta \text{ is nef} \}.\end{aligned}$$

Given a birational map $\phi: X \dashrightarrow Y$ over U , whose inverse does not contract any divisors, define

$$\mathcal{P}_Y = \{ \Delta \in \mathcal{P}_A \mid (Y, \Gamma = \phi_*\Delta) \text{ is a weak log canonical model for } (X, \Delta) \text{ over } U \},$$

and given a rational map $\psi: X \dashrightarrow Z$ over U , define

$$\mathcal{Q}_Z = \{ \Delta \in \mathcal{P}_A \mid Z \text{ is an ample model for } (X, \Delta) \text{ over } U \},$$

(cf. (3.5.4) for the definitions of weak log canonical model and ample model for (X, Δ) over U).

In nearly all applications, A will be an ample $\mathbb{Q}$ -divisor over U . In this case, we often assume that A is general in the sense that we fix a positive integer such that kA is very ample, and we assume that $A = \frac{1}{k}A'$, where $A' \in |kA|$ is very general. With this choice of A , we have

$$\mathcal{N}_A \subset \mathcal{P}_A \subset \mathcal{L}_A = \mathcal{L} + A \subset V_A = V + A.$$

The following result was first proved by Shokurov [35] assuming the existence and termination of flips:

Corollary 1.1.5. *Let $\pi: X \longrightarrow U$ be a projective morphism of normal quasi-projective varieties. Let V be a finite dimensional affine subspace of the real vector space of Weil divisors which is defined over the rationals. Suppose there is a divisor $\Delta_0 \in V$ such that $K_X + \Delta_0$ is kawamata log terminal. Let A be an ample $\mathbb{Q}$ -divisor over U .*

- (1) *There are finitely many birational maps $\phi_i: X \dashrightarrow Y_i$, $1 \leq i \leq p$, whose inverses do not contract any divisors, such that*

$$\mathcal{P}_A = \bigcup_{i=1}^p \mathcal{P}_i,$$

where each $\mathcal{P}_i = \mathcal{P}_{Y_i}$ is a rational polytope. Moreover, if $\phi: X \dashrightarrow Y$ is a log terminal model of (X, Δ) , for some $\Delta \in \mathcal{P}_A$, then $\phi = \phi_i$, some $1 \leq i \leq p$.

- (2) *There are finitely many rational maps $\psi_j: X \dashrightarrow Z_j$, $1 \leq j \leq q$ which partition $\mathcal{P}_A$ into the subsets $\mathcal{Q}_j = \mathcal{Q}_{Z_j}$.*
(3) *For every $1 \leq i \leq p$ there is a $1 \leq j \leq q$ and a morphism $f_{ij}: Y_i \rightarrow Z_j$ such that $\mathcal{P}_i \subset \overline{\mathcal{Q}}_j$.*

In particular $\mathcal{P}_A$ and each $\overline{\mathcal{Q}}_i$ are rational polytopes.

Definition 1.1.6. *Let (X, Δ) be a kawamata log terminal pair and let D be a big divisor. Suppose that $K_X + \Delta$ is not pseudo-effective. The **effective log threshold** is*

$$\sigma(X, \Delta, D) = \sup\{t \in \mathbb{R} \mid D + t(K_X + \Delta) \text{ is pseudo-effective}\}.$$

*The **Kodaira energy** is the reciprocal of the effective log threshold.*

Following ideas of Batyrev, one can easily show that:

Corollary 1.1.7. *Let (X, Δ) be a kawamata log terminal pair and let D be an ample divisor. Suppose that $K_X + \Delta$ is not pseudo-effective.*

If both $K_X + \Delta$ and D are $\mathbb{Q}$ -Cartier then the effective log threshold and the Kodaira energy are rational.

Definition 1.1.8. *Let $\pi: X \rightarrow U$ be a projective morphism of normal algebraic spaces. Let $D^\bullet = (D_1, D_2, \dots, D_k)$ be a sequence of $\mathbb{Q}$ -divisors on X . The sheaf of $\mathcal{O}_U$ -modules*

$$\mathcal{R}(\pi, D^\bullet) = \bigoplus_{m \in \mathbb{N}^k} \pi_* \mathcal{O}_X(\lfloor \sum m_i D_i \rfloor),$$

*is called the **Cox ring** associated to $D^\bullet$.*

Using (1.1.5) one can show that adjoint Cox rings are finitely generated:

Corollary 1.1.9. *Let $\pi: X \rightarrow U$ be a projective morphism of normal algebraic spaces. Fix A an ample $\mathbb{Q}$ -divisor over U . Let $\Delta_i = A + B_i$, for some $\mathbb{Q}$ -divisors $B_i \geq 0$. Assume that $D_i = K_X + \Delta_i$ is divisorially log terminal and $\mathbb{Q}$ -Cartier. Then the Cox ring*

$$\mathcal{R}(\pi, D^\bullet) = \bigoplus_{m \in \mathbb{N}^k} \pi_* \mathcal{O}_X(\lfloor \sum m_i D_i \rfloor),$$

is a finitely generated $\mathcal{O}_U$ -module.

1.2. Moduli Spaces. The main result under this heading concerns the existence of moduli spaces for varieties of general type. The quintessential results in this direction are Grothendieck's proof [12] of the existence of a moduli space for smooth curves of genus $g > 1$, and the existence of a geometrically meaningful compactification by stable curves, due to Deligne and Mumford [6]. Viehweg [39] subsequently proved the existence of the moduli space of smooth varieties with ample canonical bundle. Unfortunately this space is not projective in general and so it is interesting to find a geometrically meaningful compactification. One such candidate is the moduli space of stable pairs, introduced by Kollár and Shepherd-Barron [27]. Alexeev [1], building on their work, proved the existence of the moduli space of stable pairs for surfaces. Finally Karu [17], using the existence of a semistable reduction, was able to show that the existence of moduli space of stable pairs is only contingent on the existence of log terminal models.

We need an ad hoc:

Definition 1.2.1. *We say that a semi log canonical pair (X, Δ) can be **smoothed** to a kawamata log terminal pair of index r , if there is a flat family of pairs (X_t, Δ_t) , whose central fibre is isomorphic to (X, Δ) and whose generic fibre is a kawamata log terminal pair (Y, Γ) such that $r(K_Y + \Gamma)$ is Cartier.*

The first result gives the existence of moduli spaces for varieties of general type, and more generally kawamata log terminal pairs of log general type:

Corollary 1.2.2. *Fix a hilbert polynomial h and a positive integer r , and let $\mathcal{M}_h^{sm}$ denote the moduli functor of log pairs (X, Δ) , where*

- (1) $K_X + \Delta$ is semi-log canonical,
- (2) $K_X + \Delta$ has hilbert polynomial h ,
- (3) $K_X + \Delta$ is ample, and
- (4) (X, Δ) can be smoothed to a kawamata log terminal pair of index r .

Then $\mathcal{M}_h^{sm}$ is coarsely represented by a projective scheme M_h^{sm} .

At first sight (1.1.5) might seem a hard result to digest. For this reason, we would like to give a concrete, but non-trivial example. The moduli spaces $\overline{M}_{g,n}$ of n -pointed stable curves of genus g are probably the most intensively studied moduli spaces. In particular the problem of trying to understand the related log canonical models via the theory of moduli has attracted a lot of attention (e.g. see [11], [28] and [3]).

Corollary 1.2.3. *Let $X = \overline{M}_{g,n}$ the moduli space of curves of genus g with n marked points and let Δ_i , $1 \leq i \leq k$ denote the boundary divisors.*

Let $\Delta = \sum_i a_i \Delta_i$ be a boundary. Then $K_X + \Delta$ is log canonical and if $K_X + \Delta$ is big then there is a log canonical model $X \dashrightarrow Y$. Moreover if we fix an ample divisor A and require that $A \leq \Delta$, then the set of all log canonical models obtained this way is finite.

1.3. Fano Varieties. The next set of applications is to Fano varieties. The key observation is that given any divisor D , a small multiple of D is linearly equivalent to a divisor of the form $K_X + \Delta$, where Δ is big and $K_X + \Delta$ is kawamata log terminal.

Using this observation we can show:

Corollary 1.3.1. *Let $\pi: X \rightarrow U$ be a projective morphism of normal varieties, where U is affine. Suppose that X is $\mathbb{Q}$ -factorial, $K_X + \Delta$ is divisorially log terminal and $-(K_X + \Delta)$ is ample.*

Then X is a Mori dream space.

There are many reasons why Mori dream spaces (see [16] for the definition) are interesting. As the name might suggest, they behave very well with respect to the Minimal Model Program. Given any divisor D , one can run the D -MMP, and this ends with either a nef model, or a fibration, for which $-D$ is relatively ample, and in fact any sequence of D -flips terminates.

(1.3.1) was conjectured in [16] where it is also shown that Mori dream spaces are GIT quotients of affine varieties by a torus. Moreover the decomposition given in (1.1.5) is induced by all the possible ways of taking GIT quotients, as one varies the linearisation.

Finally, it was shown in [16] that if one has a Mori dream space, then the Cox Ring is finitely generated.

We next prove a result that naturally complements (1.2). We show that if $K_X + \Delta$ is not pseudo-effective, then we can run the MMP with scaling to get a Mori fibre space:

Corollary 1.3.2. *Let (X, Δ) be a $\mathbb{Q}$ -factorial kawamata log terminal pair. Let $\pi: X \rightarrow U$ be a projective morphism, where U is an algebraic space. Suppose that $K_X + \Delta$ is not π -pseudo-effective.*

Then we may run $f: X \dashrightarrow Y$ a $(K_X + \Delta)$ -MMP over U that ends with a Mori fibre space $g: Y \rightarrow W$.

1.4. Birational Geometry. Another immediate consequence of (1.2) is the existence of flips:

Corollary 1.4.1. *Let (X, Δ) be a kawamata log terminal pair and let $\pi: X \rightarrow Z$ be a small $(K_X + \Delta)$ -extremal contraction.*

Then the flip of π exists.

As already noted, we are unable to prove the termination of flips in general. However, using (1.1.5), we can show that any sequence of directed flips terminates, so that any sequence of flips for the MMP with scaling terminates:

Corollary 1.4.2. *Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let (X, Δ) be a $\mathbb{Q}$ -factorial kawamata log terminal pair, where $K_X + \Delta$ is $\mathbb{R}$ -Cartier. Suppose that Δ is π -big and $K_X + \Delta$ is π -pseudo-effective.*

Then we may run the directed $(K_X + \Delta)$ -MMP over U . In particular If $K_X + \Delta + tC$ is kawamata log terminal and π -nef, then we may run the $(K_X + \Delta)$ -MMP over U with scaling of C .

Another application of (1.2) is the following result concerning the factorisation of certain birational maps:

Corollary 1.4.3. *Let (X, Δ) be a log canonical pair and let $f: Y \rightarrow X$ be a log resolution. Suppose that there is a divisor Δ_0 such that $K_X + \Delta_0$ is kawamata log terminal. Let $\mathcal{E}$ be any set of exceptional divisors which satisfies the following two properties:*

- (1) *$\mathcal{E}$ contains only exceptional divisors of log discrepancy at most one, and*
- (2) *every exceptional divisor of log discrepancy one in $\mathcal{E}$ does not contain any log canonical centre.*

Then we may find a birational factorisation $\pi: W \rightarrow X$ of ϕ , such that π extracts precisely those exceptional divisors contained in $\mathcal{E}$.

The above result implies the existence of a terminal model for any kawamata log terminal pair. However it is not optimal as it does not fully address the log canonical case. Nevertheless, we are able to prove the following result (cf. [34], [25], [18]):

Corollary 1.4.4 (Inversion of Adjunction). *Let (X, Δ) be a log pair and let S be the normalisation of a component of Δ of coefficient one. Let Θ be the different.*

Then the log discrepancy of $K_S + \Theta$ is equal to the log discrepancy of $K_X + \Delta$ in a neighbourhood of S .

One of the most compelling reasons to enlarge the category of varieties to the category of algebraic spaces (equivalently Moisezon spaces, at least in the proper case) is to allow the possibility of cut and paste

operations, such as one can perform in topology. Unfortunately, it is then all too easy to construct proper smooth algebraic spaces over $\mathbb{C}$, which are not projective. In fact the appendix to [15] has two very well known examples due to Hironaka. In both examples, one exploits the fact that for two curves in a threefold which intersect in a node, the order in which one blows up the curves is important (in fact the resulting threefolds are connected by a flop).

It is then natural to wonder if this is the only way to construct such examples, in the sense that if a proper algebraic space is not projective then it must contain a rational curve. In a slightly different but related direction, it is conjectured that if a complex Kähler manifold M does not contain any rational curves then K_M is nef (see for example [33]), which would extend some of Mori's famous results from the projective case.

The following result, which was proved by Shokurov assuming the existence and termination of flips, cf. [36], gives an affirmative answer to the first conjecture and at the same time connects the two conjectures:

Corollary 1.4.5. *Let $\pi: X \longrightarrow U$ be a proper map of normal algebraic spaces, where X is $\mathbb{Q}$ -factorial.*

If $K_X + \Delta$ is divisorially log terminal and π does not contract any rational curves then π is a log terminal model. In particular π is projective and $K_X + \Delta$ is π -nef.

2. DESCRIPTION OF THE PROOF

Theorem A (Existence of log terminal models). *Let $\pi: X \longrightarrow U$ be a projective morphism of normal quasi-projective varieties, where X has dimension n . Suppose that $K_X + \Delta$ is kawamata log terminal, where Δ is big over U .*

If there exists an $\mathbb{R}$ -divisor D such that $K_X + \Delta \sim_{\mathbb{R},U} D \geq 0$, then $K_X + \Delta$ has a log terminal model over U .

Theorem B (Finiteness of models). *Let $\pi: X \longrightarrow U$ be a projective morphism of normal quasi-projective varieties, where X has dimension n . Fix A , an ample divisor over U . Suppose that $K_X + \Delta_0$ is kawamata log terminal, for some Δ_0 .*

Let $\mathcal{C} \subset \mathcal{L}_A$ be a subset with the following properties:

- (1) *For every $\Delta \in \mathcal{C} \cap \mathcal{P}_A$, there is a divisor $D \geq 0$ with $K_X + \Delta \sim_{\mathbb{R},U} D$.*
- (2) *$\mathcal{C}$ is a polytope.*

Then the set of isomorphism classes

$\{ Y \mid Y \text{ is the weak log canonical model over } U \text{ of a pair } (X, \Delta), \text{ where } \Delta \in \mathcal{C} \},$

is finite.

Theorem C (Non-vanishing theorem). *Let $\pi: X \longrightarrow U$ be a projective morphism of normal quasi-projective varieties, where X has dimension n . Suppose that $K_X + \Delta$ is kawamata log terminal, where Δ is big over U .*

If $K_X + \Delta$ is π -pseudo-effective, then there exists an $\mathbb{R}$ -divisor D such that $K_X + \Delta \sim_{\mathbb{R}, U} D \geq 0$.

The proof of Theorem A, Theorem B and Theorem C proceeds by induction:

- Theorem B_{n-1} and Theorem C_{n-1} imply Theorem A_n , cf. (5.9).
- Theorem B_{n-1} and Theorem A_n imply Theorem B_n , cf. (6.3).
- Theorem A_n and Theorem B_n imply Theorem C_n , cf. (7.4).

2.1. Sketch of the proof. To help the reader navigate through the technical problems which arise naturally when trying to prove (1.2), we review a natural approach to proving that the canonical ring

$$R(X, K_X) = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(mK_X)),$$

of a smooth projective variety X of general type is finitely generated. Even though we do not directly follow this method to prove the existence of log terminal models, instead using ideas from the MMP, many of the difficulties which arise in our approach are mirrored in trying to prove finite generation directly.

A very natural way to proceed is to pick a divisor $B \in |kK_X|$, whose existence is guaranteed as we assuming that K_X is big, and then try to restrict to B . One obtains an exact sequence

$$0 \longrightarrow H^0(X, \mathcal{O}_X((m-k)K_X)) \longrightarrow H^0(X, \mathcal{O}_X(mK_X)) \longrightarrow H^0(B, \mathcal{O}_B(mK_B)),$$

and it is easy to see that it suffices to prove that the restricted algebra, the image of the map

$$H^0(X, \mathcal{O}_X(mK_X)) \longrightarrow H^0(B, \mathcal{O}_B(mK_B)),$$

is finitely generated. Various problems arise at this point. First B is neither smooth nor even reduced (which, for example, means that the symbol K_B is only formally defined; strictly speaking we ought to work with ω_B). It is natural then to pass to a log resolution, so that the support D of B has simple normal crossings, and to replace B by D . The second problem is that the kernel of the map

$$H^0(X, \mathcal{O}_X(mK_X)) \longrightarrow H^0(D, \mathcal{O}_D(mK_D)),$$

no longer has any obvious connection with $H^0(X, \mathcal{O}_X((m - k)K_X))$, so that even if we knew that the new restricted algebra were finitely generated, it is not immediate that this is enough. Another significant problem is to identify the restricted algebra as a subalgebra of $\bigoplus_{m \in \mathbb{N}} H^0(D, \mathcal{O}_D(mK_D))$, since it is only the latter that we can handle by induction. Yet another problem is that if C is a component of D , it is no longer the case that C is of general type, so that we need a more general induction. In this case the most significant problem to deal with is that even if K_C is pseudo-effective, it is not clear that the linear system $|kK_C|$ is non-empty for any $k > 0$. Finally, even though this aspect of the problem may not be apparent from the description above, in practice it seems as though we need to work with infinitely many different values of k and hence $B = B_k$, which entails working with infinitely many different birational models of X (since for every different value of k , one needs to resolve the singularities of D).

We now review our approach to the proof of (1.2). As is clear from the plan of the proof given in the previous subsection, the proof of (1.2) is by induction on the dimension and the proof is split into three parts. However instead of proving that the canonical ring is finitely generated, we try to construct a minimal model for X . By the main result of [13] the main issue is to prove that the MMP terminates, which means that we must show that we only need finitely many flips.

As in the scheme of the proof of finite generation sketched above, the first step is to pick $B \in |kK_X|$, and to pass to a log resolution of the support D of B . By way of induction we want to work with $K_X + D$ rather than K_X . As before this is tricky since a minimal model for $K_X + D$ is not the same as a minimal model for K_X . In other words having added D , we really want to subtract it as well. The trick however is to first add D , construct a minimal model for $K_X + D$ and then subtract D (almost literally component by component). This is the key step, to show that Theorem B _{$n-1$} and Theorem C _{$n-1$} imply Theorem A _{n} . This part of the proof splits naturally into two parts. First we have to prove that we may run the relevant minimal model programs, see §4 and then we have to prove this does indeed construct a minimal model for $K_X + \Delta$, see §5.

To gain intuition for how this part of the proof works, let us consider a series of successively harder cases. First suppose that $D = S$ is irreducible. In this case it is clear that K_X is nef iff $K_X + S$ is nef and in fact a log terminal model for K_X is the same as a log terminal model for $K_X + S$. Consider running the $(K_X + S)$ -MMP. Then every step of this MMP is a step of the K_X -MMP and vice-versa. Suppose that we have a $(K_X + S)$ -extremal ray R . Let $\pi: X \rightarrow Z$ be the corresponding

contraction. Then $S \cdot R < 0$, so that every curve Σ contracted by π must be contained in S . If π is a divisorial contraction, then π must contract S and $K_Y \sim_{\mathbb{Q}} 0$, so that Y is a log terminal model. Otherwise by induction and the main result of [13], we can construct the flip of π , $\phi: X \dashrightarrow Y$. Consider the restriction $\psi: S \dashrightarrow T$ of ϕ to S , where T is the strict transform of S . Since log discrepancies increase under flips and S is irreducible, ψ^{-1} cannot contract any divisors. After finitely many flips, we may therefore assume that ψ does not contract any divisors, since the Picard number of S cannot keep dropping. Consider what happens if we restrict to S . By adjunction, we have

$$(K_X + S)|_S = K_S.$$

Thus $\psi: S \dashrightarrow T$ is K_S -negative. We have to show that this cannot happen infinitely often. If we knew that every sequence of flips on S terminates, then we would be done. In fact this is how special termination works. Unfortunately we cannot prove that every sequence of flips terminates on S , so that we have to do something slightly different. Instead we throw in an auxiliary ample divisor H on X , and consider $K_X + S + tH$, where t is a positive real number. If t is large enough then $K_X + S + tH$ is ample. Decreasing t , we may assume that there is an extremal ray R such that $(K_X + S + tH) \cdot R = 0$. If $t = 0$, then $K_X + S$ is nef and we are done. Otherwise $(K_X + S) \cdot R < 0$, so that we are still running a $(K_X + S)$ -MMP, but with the additional restriction that $K_X + S + tH$ is nef and trivial on any ray we contract. Let $G = H|_S$. Then $K_S + tG$ is nef and so is $K_T + tG'$, where $G' = \psi_* G$. In this case $K_T + tG'$ is a weak log canonical model for $K_S + tG$ (it is not a log terminal model, both because ϕ might contract divisors on which $K_S + tG$ is trivial and more importantly because T need not be $\mathbb{Q}$ -factorial). In this case we are then done, by finiteness of weak log canonical models for (S, tG) , where $t \in [0, 1]$.

Now we consider a slightly more general case, where $K_X \sim sS = s \sum S_i$, for some natural number s . The main problem here is that not every component S_i of S is of general type, and so the induction on dimension is a little more involved. On the other hand, since K_X is big, we can always pick $\Delta \sim_{\mathbb{Q}} \epsilon K_X$, for some $\epsilon > 0$, such that $\Delta = A + B$, where A is ample and $B \geq 0$ and $K_X + \Delta$ is kawamata log terminal. Note that a log terminal model for K_X is the same as a log terminal model for $K_X + A + B$. Assuming that no component of Δ is a component of S , a log terminal model for $K_X + A + B$ is the same as a log terminal model for $K_X + S + A + B$. As before if R is an extremal ray which is $(K_X + S + A + B)$ -negative, then $S \cdot R < 0$, so that $S_i \cdot R < 0$ for some component S_i of S . As before this guarantees existence of flips.

Now if the $(K_X + S + A + B)$ -MMP does not terminate, then we must have an infinite sequence of flips which must intersect one component S_i of S infinitely often. Using the ample divisor A to tie break, and since the only issue is termination, we may assume that S is irreducible. The first added complication is that if $\psi: S \dashrightarrow T$ is the restriction of a flip $\phi: X \dashrightarrow Y$ to S , then ψ^{-1} might contract some divisors. However it is a standard result that this can only happen finitely many times. As before, we run a $(K_X + S + A + B)$ -MMP, with scaling of H , so that we only consider $(K_X + S + A + B)$ -extremal rays R , which are $(K_X + S + A + B + tH)$ -trivial, where $K_X + S + A + B + tH$ is nef. If we restrict to S , then we have

$$(K_X + S + A + B)|_S = K_S + C + D,$$

where $C = A|_S$ is ample and $D = B|_S \geq 0$. As $K_X + S + A + B$ is purely log terminal, $K_S + C + D$ is kawamata log terminal, so that we have exactly the right hypotheses to apply induction, and we can use finiteness of weak log canonical models for S to finish, as before.

Now consider an even more general case, $K_X \sim_{\mathbb{Q}} \sum s_i S_i$, where $s_1, s_2, \dots, s_k$ are not necessarily equal. Note that $K_X + S + A + B \sim_{\mathbb{Q}} L \geq 0$, where the support of L is equal to S . Thus if $(K_X + S + A + B) \cdot R < 0$, then $L \cdot R < 0$, so that $S_i \cdot R < 0$, for some component of S_i of S . Thus, by the same argument as before, we may construct a log terminal model for $K_X + S + A + B$. The main new difficulty is that a log terminal model for $K_X + S + A + B$ is not necessarily a log terminal model for K_X . However it is not too hard to get around this problem. For k sufficiently large, we may decompose S as $M + F$, where M is the mobile part of $|kK_X|$ for some k (so that M is either irreducible or empty) and the components of F are all components of the stable base locus. Then replace X by a log resolution, so that the support of $S + A + B$ has global normal crossings. Running the $(K_X + S + A + B)$ -MMP with scaling of H , as before, we may assume that $K_X + S + A + B$ is nef. At this stage we want to subtract M . So now we run the $(K_X + F + A + B)$ -MMP, with scaling of M . Suppose that R is a $(K_X + F + A + B)$ -negative ray. By definition of the MMP with scaling, $M \cdot R > 0$. But then $F_i \cdot R < 0$ for some component F_i of F . At the end, we have that $K_X + F + A + B$ is nef. Since A is big, by the base point free theorem, $K_X + F + A + B$ is therefore semiample. Since every component of F is part of the stable base locus of $K_X + A + B$, it is also part of the stable base locus of $K_X + F + A + B$, and it must be the case that F is empty, that is, it has been contracted by the two MMPs we have just run. Finally, we need to observe that since we only contracted divisors on which $K_X + A + B$ is negative, even

though we did not run a $K_X + A + B$ -MMP, in fact we still constructed a log terminal model for $K_X + A + B$, which is the same a log terminal model for K_X .

Now we consider the most general case. In this case we need to start with $K_X + \Delta$, where $\Delta = A + B$ is an $\mathbb{R}$ -divisor (we will see later why it is necessary to work at this level of generality). In this case $K_X + \Delta \sim_{\mathbb{R}} \sum s_i S_i$. Assume that B and S have no common components (we only assume this to simplify the notation). As before the first step is to run a MMP so that $K_X + S + A + B$ is nef. The argument is almost exactly the same as before. Unfortunately if we try to write $S = M + F$, since we are now dealing with $\mathbb{R}$ -divisors, we can no longer assume that M is irreducible. The best we can arrange for is that every component of M is mobile. With this choice of M , at least it is the case that no component can ever be contracted, by running a MMP. So now the process of subtracting M is much more delicate. In fact we need to subtract each component of M one at a time. Then we can finish off as before. The details are contained in §5.

Now we explain the rest of the proof. In terms of induction, we need to prove finiteness of weak log canonical models. We fix an ample divisor A and work with divisors of the form $K_X + \Delta = K_X + A + B$, where the coefficients of B are variable. For ease of exposition, we assume that the support of A and B have global normal crossings, so that $K_X + \Delta = K_X + A + \sum b_i B_i$ is log canonical iff $0 \leq b_i \leq 1$ for all i . The key point is that we allow the coefficients of B to be real numbers, so that the set of all possible choices of coefficients $[0, 1]^k$ is a compact subset of $\mathbb{R}^k$. Thus we may check finiteness locally. In fact since A is ample, we can always perturb the coefficients of B so that none of the coefficients are equal to one or zero and so we may even assume that $K_X + \Delta$ is kawamata log terminal.

Observe that we are certainly free to add components to B (formally we add components with coefficient zero and then perturb so that their coefficients are non-zero). In particular we may assume that B is the support of an ample divisor and so working on the weak log canonical model, we may assume that we have a log canonical model for a perturbed divisor. Thus it suffices to prove that there are only finitely many log canonical models. Since the log canonical model is determined by any log terminal model, it suffices to prove that we can find a cover of $[0, 1]^k$ by finitely many log terminal models. By compactness, it suffices to do this locally.

So pick $b \in [0, 1]^k$. There are two cases. If $K_X + \Delta$ is not pseudo-effective, then $K_X + A + B'$ is not pseudo-effective, for B' in a neighbourhood of B , and there are no weak log canonical models at all.

Otherwise we may assume that $K_X + \Delta$ is pseudo-effective. Suppose that we know that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$. Then we know that there is a log terminal model $\phi: X \dashrightarrow Y$. Replacing (X, Δ) by $(Y, \Gamma = \phi_* \Delta)$, we may assume that $K_X + \Delta$ is nef. By the base point free theorem it is semiample. Let $X \dashrightarrow Z$ be the corresponding morphism. The key observation is that locally about Δ , any log terminal model over Z is an absolute log terminal model. Working over Z , we may assume that $K_X + \Delta$ is numerically trivial. In this case the problem of finding a log terminal model for $K_X + \Delta'$ only depends on the line segment spanned by Δ and Δ' . Working in a small box about Δ , we are then reduced to finding a log terminal model on the boundary of the box and we are done by induction on the dimension of the affine space containing B . Note that in practice, we need to work in slightly more generality than we have indicated; first we need to work in the relative setting and secondly we need to work with an arbitrary affine space containing B (and not just the space spanned by the components of B). This poses no significant problem. This explains the implication Theorem B_{n-1} and Theorem A_n imply Theorem B_n . The details are contained in §6.

Finally we need to explain how to prove that if $K_X + \Delta = K_X + A + B$ is pseudo-effective, then $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$. The idea is to mimic the proof of the non-vanishing theorem. As in the proof of the non-vanishing theorem and following the work of Nakayama, there are two cases. In the first case, for any ample divisor H ,

$$h^0(X, \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor + H))$$

is a bounded function of m . It is not hard to prove directly that $K_X + \Delta$ has a log terminal model and we are done by the base point free theorem.

In the second case we construct a log canonical centre for $m(K_X + \Delta) + H$, when m is sufficiently large. Passing to a log resolution, and using standard arguments, we are reduced to the case when $K_X + \Delta = K_X + S + A + B$, where S is irreducible and $(K_X + \Delta)|_S$ is pseudo-effective, and the support of Δ has global normal crossings. Suppose first that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. We may write

$$(K_X + S + A + B)|_S = K_S + C + D,$$

where C is ample and $D \geq 0$. By induction we know that there is a positive integer m such that $h^0(S, \mathcal{O}_S(m(K_S + C + D))) > 0$. To lift sections, we need to know that $h^1(X, \mathcal{O}_X(m(K_X + S + A + B) - S)) = 0$. Now

$$m(K_X + S + A + B) - K_X - S = (m-1)(K_X + S + A + B) + A = (m-1)(K_X + \Delta + \frac{1}{m-1}A).$$

As $K_X + \Delta + A/(m-1)$ is big, we can construct a log terminal model $\phi: Y \dashrightarrow X$ for $K_X + \Delta + A/(m-1)$, and running this argument on Y , the required vanishing holds by Kawamata-Viehweg vanishing. In the general case, $K_X + S + A + B$ is an $\mathbb{R}$ -divisor. The argument is now a little more delicate as $h^0(S, \mathcal{O}_S(m(K_S + C + D)))$ does not make sense. We need to approximate $K_S + C + D$ by rational divisors, which we can do by induction. But then it is not so clear how to choose m . In practice we need to prove that the log terminal model Y constructed above does not depend on m , at least locally in a neighbourhood of T , the strict transform of S , and then the result follows by Diophantine approximation. This explains the implication Theorem A_n and Theorem B_n imply Theorem C_n . The details are in §7.

2.2. Standard conjectures of the MMP. Having sketched the proof of (1.2), we should point out the main obstruction to extending these ideas to the case when X is not of general type. The main issue seems to be the implication K_X pseudo-effective implies $\kappa(X, K_X) \geq 0$. In other words we need:

Conjecture 2.1. *Let (X, Δ) be a projective kawamata log terminal pair.*

If $K_X + \Delta$ is pseudo-effective then $\kappa(X, K_X + \Delta) \geq 0$.

We also probably need

Conjecture 2.2. *Let (X, Δ) be a projective kawamata log terminal pair.*

If the function

$$h^0(X, \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor + H))$$

is not a bounded function of m , for some ample divisor H , then $\kappa(X, K_X + \Delta) \geq 1$.

In fact taken together, using the methods of this paper, together with some results of Kawamata, [19] and [20], (2.1) and (2.2), would seem to imply one of the main outstanding conjectures of higher dimensional geometry:

Conjecture 2.3 (Abundance). *Let (X, Δ) be a kawamata log terminal pair.*

If $K_X + \Delta$ is nef then it is semiample.

It also seems worth pointing out that the other remaining conjecture is:

Conjecture 2.4 (Borisov-Alexeev-Borisov). *Fix a positive integer n and a positive real number $\epsilon > 0$.*

Then the set of varieties X such that $K_X + \Delta$ has log discrepancy at least ϵ and $-(K_X + \Delta)$ is ample, forms a bounded family.

Finally, it seems worthwhile pointing out that a seemingly innocuous generalisation of (1.2):

Conjecture 2.5. *Let (X, Δ) be a log canonical pair.*

If $K_X + \Delta$ is big, then (X, Δ) has a log canonical model.

would imply (2.3). Note that the only difference is that we have generalised from kawamata log terminal to log canonical.

We sketch a possible argument. Pick (X, Δ) a projective kawamata log terminal pair. Let X' be the cone over X with vertex p , and let $Y \rightarrow X$ be the blow up of X' at p . Then the exceptional divisor E is a copy of X . Let Δ' be the cone over Δ and let $\Gamma = \Delta_0 + E + D$, where Δ_0 is the strict transform of Δ' and D is the strict transform on Y of any ample divisor H on X' . Now if H is sufficiently ample, then $K_{X'} + \Delta' + H$ is ample and $K_Y + \Gamma$ is big. On the other hand, if we write

$$(K_Y + \Gamma)|_E = K_E + \Psi,$$

then the pair (E, Ψ) is naturally isomorphic to (X, Δ) . Let $\phi: Y \dashrightarrow W$ be a log canonical model for $(Y, \Gamma + \epsilon H)$, where H is ample and $\epsilon > 0$ is sufficiently small and let $W \rightarrow Z$ be a log canonical model. Then ϕ is an isomorphism outside E . It is not hard to see that the restriction $\psi: E \dashrightarrow F$ of ϕ to E , where F is the strict transform of E , is a log terminal model for (X, Δ) , and the induced morphism $F \rightarrow F'$, where F' is the image of F in Z is the map given by abundance.

3. PRELIMINARY RESULTS

In this section we collect together some definitions and results that will be needed for the proof of (1.4.2).

3.1. Notation and conventions. We work over the field of complex numbers $\mathbb{C}$. We say that two $\mathbb{Q}$ -divisors D_1, D_2 are $\mathbb{Q}$ -linearly equivalent ($D_1 \sim_{\mathbb{Q}} D_2$) if there exists an integer $m > 0$ such that mD_i are linearly equivalent.

We recall some definitions involving divisors with real coefficients:

Definition 3.1.1. *Let $\pi: X \rightarrow U$ be a proper morphism of normal algebraic spaces.*

- (1) *An **$\mathbb{R}$ -Weil divisor** D on X is an $\mathbb{R}$ -linear combination of prime divisors.*

- (2) Two $\mathbb{R}$ -divisors D and D' are **$\mathbb{R}$ -linearly equivalent over U** , denoted $D \sim_{\mathbb{R},U} D'$, if their difference is an $\mathbb{R}$ -linear combination of principal divisors and a divisor pulled back from U .
- (3) An **$\mathbb{R}$ -Cartier divisor** D is an $\mathbb{R}$ -linear combination of Cartier divisors.
- (4) An $\mathbb{R}$ -Cartier divisor D is **ample over U** if it is $\mathbb{R}$ -linearly equivalent to a positive linear combination of ample (in the usual sense) divisors over U .
- (5) A $\mathbb{R}$ -Cartier divisor D on X is **nef over U** if $D \cdot C \geq 0$ for any curve $C \subset X$, contracted by π .
- (6) An $\mathbb{R}$ -divisor D is **effective** if it is a positive real linear combination of prime divisors.
- (7) An $\mathbb{R}$ -divisor D is **big over U** if

$$\limsup \frac{h^0(F, \mathcal{O}_F(\lfloor mD \rfloor))}{m^{\dim D}} > 0,$$

for the fibre F over any generic point of U .

- (8) An $\mathbb{R}$ -Cartier divisor D is **semiample over U** if there is a morphism $\pi: X \rightarrow Y$ over U such that D is $\mathbb{R}$ -linearly equivalent to the pullback of an ample divisor over U .

Note that the group of Weil divisors with rational or real coefficients forms a vector space, with a canonical basis given by the prime divisors. Given an $\mathbb{R}$ -divisor, $\|D\|$ denotes the sup norm with respect to this basis. If A and B are two $\mathbb{R}$ -divisors, then we let $[A, B]$ denote the line segment

$$\{ \lambda A + \mu B \mid \lambda + \mu = 1, \lambda \geq 0, \mu > 0 \}.$$

Given an $\mathbb{R}$ -divisor D and a subvariety Z which is not contained in the singular locus of X , $\text{mult}_Z D$ denotes the multiplicity of D at the generic point of Z . If $Z = E$ is a prime divisor this is the coefficient of E in D .

A *log pair* (X, Δ) is a normal variety X and an effective $\mathbb{R}$ -Weil divisor Δ such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier. We say that a log pair (X, Δ) is *log smooth*, if X is smooth and the support of Δ is a divisor with global normal crossings. A projective morphism $g: Y \rightarrow X$ is a *log resolution* of the pair (X, Δ) if Y is smooth and $g^{-1}(\Delta)$ union the exceptional set of g is a divisor with normal crossings support. We write

$$K_Y + \Gamma = g^*(K_X + \Delta),$$

and $\Gamma = \sum a_i \Gamma_i$ where Γ_i are distinct reduced irreducible divisors. The log discrepancy of Γ_i is $1 - a_i$. The *locus of log canonical singularities of the pair (X, Δ)* , denoted $\text{LCS}(X, \Delta)$, is equal to the image of

those components of Γ of coefficient at least one (equivalently log discrepancy at most zero). The pair (X, Δ) is *kawamata log terminal* if for every (equivalently for one) log resolution $g: Y \rightarrow X$ as above, the coefficients of Γ are strictly less than one, that is $a_i < 1$ for all i . Equivalently, the pair (X, Δ) is kawamata log terminal if the locus of log canonical singularities is empty. We say that the pair (X, Δ) is *purely log terminal* if the log discrepancy of any exceptional divisor is greater than zero. We say that the pair (X, Δ) is *divisorially log terminal* if there is a log resolution such that the log discrepancy of every exceptional divisor is greater than zero.

We will also often write

$$K_Y + \Gamma = g^*(K_X + \Delta) + E,$$

where Γ and E are effective, with no common components, $g_*\Gamma = \Delta$ and E is g -exceptional. Note that this decomposition is unique.

3.2. Algebraic spaces. We will need a minimal amount of standard material concerning algebraic spaces. Unless otherwise stated all algebraic spaces are reduced, separated and of finite type over $\mathbb{C}$.

Lemma 3.2.1. *Let $\pi: X \rightarrow U$ be a proper morphism of normal algebraic spaces. Suppose that there is a finite cover $f: V \rightarrow U$ such that V is a quasi-projective variety. Let D be a $\mathbb{R}$ -Cartier divisor on X and let D' be its restriction to the generic fibre of π .*

If $D' \sim_{\mathbb{R}} B' \geq 0$ for some $\mathbb{R}$ -divisor B' on the generic fibre of π , then there is a divisor B on X such that $D \sim_{\mathbb{R}, U} B \geq 0$

Proof. First assume that f is the identity.

Taking the closure of the generic points of B' , we may assume that there is an effective $\mathbb{R}$ -divisor B_1 on X such that the restriction of B_1 to the generic fibre is B' . As

$$D' - B' \sim_{\mathbb{R}} 0,$$

it follows that there is an open subset U_1 of U , such that

$$(D - B_1)|_{V_1} \sim_{\mathbb{R}} 0,$$

where V_1 is the inverse image of U_1 . But then there is a divisor G on X such that

$$D - B_1 \sim_{\mathbb{R}} G,$$

where $Z = \pi(\text{Supp } G)$ is a proper closed subset. As U is quasi-projective, there is an ample divisor H on U which contains Z . Possibly rescaling, we may assume that $F = \pi^*H \geq -G$. But then

$$D \sim_{\mathbb{R}} (B_1 + F + G) - F,$$

so that

$$D \sim_{\mathbb{R}, U} (B_1 + F + G) \geq 0.$$

Now consider the general case. Passing to a finite cover of V , we may assume that f is Galois, with Galois group G and that V is normal. Let

$$\begin{array}{ccc} Y & \xrightarrow{g} & X \\ \psi \downarrow & & \downarrow \pi \\ V & \xrightarrow{f} & U, \end{array}$$

be the normalisation of the fibre square. By what we have already proved, we may find C on Y such that $g^*D \sim_{\mathbb{R}, U} C \geq 0$. Taking the average over G we may assume that C is G -invariant. But then there is a divisor B on X such that $C = g^*B$. We may write

$$g^*D = g^*B + \sum r_i(g_i) + \psi^*L,$$

where $r_1, r_2, \dots, r_k$ are real numbers, $g_1, g_2, \dots, g_k$ are rational functions on Y and L is an $\mathbb{R}$ -Cartier divisor on V . Averaging over G again, we may assume that $g_i = g^*f_i$, for rational functions $f_1, f_2, \dots, f_k$ on X and that $L = f^*M$, for some $\mathbb{R}$ -Cartier divisor on U . But then

$$D = B + \sum r_i(f_i) + \phi^*M. \quad \square$$

3.3. Zariski Decomposition. We will need some definitions and results from [31]:

Definition-Lemma 3.3.1. *Let X be a smooth projective variety, B be a big $\mathbb{R}$ -divisor and let C be a prime divisor. Let*

$$\sigma_C(B) = \inf \{ \text{mult}_C(B') \mid B' \sim_{\mathbb{Q}} B, B' \geq 0 \}.$$

Then σ_C is a continuous function on the cone of big divisors.

Now let D be any pseudo-effective $\mathbb{R}$ -divisor and let A be any ample $\mathbb{Q}$ -divisor. Let

$$\sigma_C(D) = \lim_{\epsilon \rightarrow 0} \sigma_C(D + \epsilon A).$$

Then $\sigma_C(D)$ exists and is independent of the choice of A .

There are only finitely many prime divisors C such that $\sigma_C(D) > 0$. Set $N_\sigma(D) = \sum_C \sigma_C(D)C$, then $N_\sigma(D)$ is determined by $c_1(D)$.

Proof. The first statement is (3) of (2.1.1) of [31] and the rest is (2.1.6) of [31]. $\square$

Proposition 3.3.2. *Let X be a smooth projective variety and let D be a pseudo-effective $\mathbb{R}$ -divisor. Let B be any big $\mathbb{R}$ -divisor.*

If D is not numerically equivalent to $N_\sigma(D)$, then there is a positive integer k and a positive rational number β such that

$$h^0(X, \mathcal{O}_X(\lfloor mD \rfloor + \lfloor kB \rfloor)) > \beta m, \quad \text{for all } m \gg 0.$$

Proof. Let A be any integral divisor. Then we may find a positive integer k such that

$$h^0(X, \mathcal{O}_X(\lfloor kB \rfloor - A)) \geq 0.$$

Thus it suffices to exhibit an ample divisor A and a positive rational number β such that

$$h^0(X, \mathcal{O}_X(\lfloor mD \rfloor + A)) > \beta m \quad \text{for all } m \gg 0.$$

Replacing D by $D - N_\sigma(D)$, we may assume that $N_\sigma(D) = 0$. Now apply (6.1.12) of [31]. $\square$

3.4. Log pairs. We recall some basic facts and standard manipulations for log pairs. First the following result, which is well known in the case of $\mathbb{Q}$ -divisors and is due to Kawamata and Shokurov:

Theorem 3.4.1 (Base Point Free Theorem). *Let (X, Δ) be a $\mathbb{Q}$ -factorial kawamata log terminal pair, where Δ is a $\mathbb{R}$ -divisor. Let $f: X \rightarrow U$ be a projective morphism of normal algebraic spaces, and let D be a nef $\mathbb{R}$ -divisor over U , such that $aD - (K_X + \Delta)$ is nef and big, for some positive real number a .*

Then D is semiample over U .

Proof. As the property that D is semiample over U is local in the étale topology, we may assume that U is affine. This case is (7.1) of [13] (for example). $\square$

We also recall some basic facts about adjunction and the different, see [25] for more details. Let (X, Δ) be a log canonical pair, and let S be a normal component of $\lfloor \Delta \rfloor$ of coefficient one. Then there is a divisor Θ on S such that

$$(K_X + \Delta)|_S = K_S + \Theta.$$

If (X, Δ) is divisorially log terminal then so is $K_S + \Theta$ and if (X, Δ) is purely log terminal then $K_S + \Theta$ is kawamata log terminal. The pair (S, Θ) only depends on a neighbourhood of S in X .

Lemma 3.4.2. *Let $\pi: X \rightarrow C$ be a flat projective morphism from a normal variety X to the germ of a nonsingular curve $0 \in C$ such that (X, Δ) is a pair where $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. If the special fibre*

(X_0, Δ_0) is semi log canonical, then the general fibre (X_t, Δ_t) is log canonical.

Proof. It suffices to prove that X is log canonical in a neighbourhood of X_0 . This follows from [18] and the definition of semi log canonical. $\square$

3.5. Types of model.

Definition 3.5.1. Let $\phi: X \dashrightarrow Y$ be a birational map, whose inverse does not contract any divisors, and let D be a $\mathbb{R}$ -Cartier divisor such that $D' = \phi_* D$ is also $\mathbb{R}$ -Cartier. We say that ϕ is **D -non-positive** (respectively **D -negative**) if for some common resolution $p: W \rightarrow X$ and $q: W \rightarrow Y$, we may write

$$p^* D = q^* D' + E,$$

where $E \geq 0$ is q -exceptional (respectively the support of E contains the strict transform of the ϕ -exceptional locus).

We will often use the following well-known:

Lemma 3.5.2 (Negativity of contraction). Let $\pi: Y \rightarrow X$ be a birational morphism.

- (1) If Y is smooth and $E > 0$ is an exceptional $\mathbb{R}$ -divisor, then there is a component F of E which is covered by curves Σ such that $E \cdot \Sigma < 0$.
- (2) If $\pi^* L = M + G + E$, where L is an $\mathbb{R}$ -Cartier divisor on X , M is a π -nef $\mathbb{R}$ -Cartier divisor on Y , $G \geq 0$, E is π -exceptional, and G and E have no common components, then $E \geq 0$. Further if E_i is a component of E such that there is a component E_j of E with the same centre on X as E_i , with the restriction of M to E_j not numerically π -trivial, then the coefficient of E_i is strictly positive.

Proof. Cutting by hyperplanes in X , we reduce to the case when X is a surface, in which case (1) reduces to the Hodge Index Theorem. (2) follows easily from (1), see for example (2.19) of [25]. $\square$

Lemma 3.5.3. Let $\phi: X \dashrightarrow Y$ be a birational map, whose inverse does not contract any divisors, and let D be a $\mathbb{R}$ -Cartier divisor such that $D' = \phi_* D$ is nef.

Then ϕ is D -non-positive (respectively D -negative) if we may write

$$D = \phi^* D' + E,$$

where $E \geq 0$ is ϕ -exceptional (respectively the support of E is equal to the ϕ -exceptional locus).

Proof. Easy consequence of (3.5.2). $\square$

Definition 3.5.4. Let $\pi: X \longrightarrow U$ be a projective morphism of normal algebraic spaces. Suppose that $K_X + \Delta$ is log canonical and let $\phi: X \dashrightarrow Y$ be a birational map over U such that ϕ^{-1} does not contract any divisors. Set $\Gamma = \phi_*\Delta$.

- Y is a **weak log canonical model** for $K_X + \Delta$ over U if ϕ is $(K_X + \Delta)$ -non-positive and $K_Y + \Gamma$ is nef over U .
- Y is a **log canonical model** for $K_X + \Delta$ if ϕ is $(K_X + \Delta)$ -non-positive and $K_Y + \Gamma$ is ample over U .
- Y is a **log terminal model** for $K_X + \Delta$ if ϕ is $(K_X + \Delta)$ -negative, $K_Y + \Gamma$ is divisorially log terminal and nef over U , and Y is $\mathbb{Q}$ -factorial.

Now suppose that $\psi: X \dashrightarrow Z$ is a rational map.

- Z is an **ample model** for $K_X + \Delta$ over U if there is a log terminal model $\phi: X \dashrightarrow Y$ for $K_X + \Delta$ over U , a morphism $f: Y \longrightarrow Z$ over U and a divisor H on Z , which is ample over U , such that $K_Y + \Gamma = f^*H$, where $\Gamma = \phi_*\Delta$.

Lemma 3.5.5. Let $\pi: X \longrightarrow U$ be a proper morphism of normal algebraic spaces. Let (X, Δ) be a kawamata log terminal pair. Let $f: Z \longrightarrow X$ be any log resolution of (X, Δ) and suppose that we write

$$K_Z + \Phi_0 = f^*(K_X + \Delta) + E,$$

where Φ_0 and E are effective, with no common components, $f_*\Phi_0 = \Delta$ and E is exceptional. Let F be any divisor whose support is equal to the exceptional locus.

Then we may find $\eta > 0$ such that if $\Phi = \Phi_0 + \eta F$ then

- $f_*\Phi = \Delta$,
- $K_Z + \Phi$ is kawamata log terminal,
- if Δ is big over U then so is Φ , and
- the log terminal models (respectively weak log canonical models) over U of $K_X + \Delta$ and the log terminal models (respectively weak log canonical models) over U of $K_Z + \Phi$ are the same.

Proof. Everything is clear, apart from the fact that if $\phi: Z \dashrightarrow W$ is a log terminal model (respectively weak log canonical model) over U of $K_Z + \Phi$ then it is a log terminal model (respectively weak log canonical model) over U of $K_X + \Delta$.

Let $\psi: X \dashrightarrow W$ be the induced birational map and set $\Psi = \phi_*\Phi$. By what we have already observed, possibly blowing up more, we may assume that ϕ is a morphism. By assumption if we write

$$K_Z + \Phi = \phi^*(K_W + \Psi) + G,$$

then $G > 0$ and the support of G is the full ϕ -exceptional locus (respectively G is exceptional). Thus

$$f^*(K_X + \Delta) + E + \eta F = \phi^*(K_W + \Psi) + G.$$

By negativity of contraction $G - E - \eta F \geq 0$, so that in particular ϕ must contract every f -exceptional divisor and ψ^{-1} does not contract any divisors. But then, ϕ is a log terminal model (respectively weak log canonical model) over U by (3.5.3). $\square$

Lemma 3.5.6. *Let $\pi: X \rightarrow U$ be a projective morphism of normal algebraic spaces. Suppose that $K_X + \Delta$ is log terminal.*

Then the ample model of $K_X + \Delta$ over U is unique, if it exists at all.

Proof. Let Z_1 and Z_2 be ample models for $K_X + \Delta$ over U . By definition we may find weak log terminal models $\phi_i: X \dashrightarrow Y_i$ for $K_X + \Delta$ over U and morphisms $f_i: Y_i \rightarrow Z_i$ over U such that, $K_{Y_i} + \Gamma_i = f_i^* H_i$, where H_i is an ample divisor over U and $\Gamma_i = \phi_{i*} \Delta$, for $i = 1$ and 2 .

Suppose that $g: W \rightarrow X$ resolves the indeterminacy of ϕ_i , for $i = 1$ and 2 . Let $\Psi \geq 0$ be the divisor on W whose existence is guaranteed by (3.5.5). Then (Y_i, Γ_i) is also a log terminal model for $K_W + \Psi$, for $i = 1$ and 2 .

Thus replacing (X, Δ) by (W, Ψ) we may assume that ϕ_i is a morphism over U . In particular, there are effective divisors E_i , exceptional for ϕ_i , such that

$$K_X + \Delta = \phi_i^*(K_{Y_i} + \Gamma_i) + E_i,$$

for $i = 1$ and 2 . As $K_{Y_i} + \Gamma_i$ is nef, for $i = 1$ and 2 , negativity of contraction implies that $E_1 = E_2$. In particular $g_1^* H_1 = g_2^* H_2$, where $g_i = f_i \circ \phi_i$ and so $Z_1 \simeq Z_2$. $\square$

3.6. Convex geometry and Diophantine Approximation.

Definition 3.6.1. *Let V be a real affine space. A **polytope** $\mathcal{P}$ in V is the convex hull of any finite set X of points.*

*If $\mathcal{C}$ is a convex subset of V then we say that $v \in \mathcal{C}$ is an **extreme point** if whenever*

$$v = \sum r_i v_i,$$

where $r_1, r_2, \dots, r_k$ are real numbers such that $\sum r_i = 1$ and $v_1, v_2, \dots, v_k$ belong to $\mathcal{C}$ then either $r_i \leq 0$ or $v = v_i$ for some i .

*We say that V is **defined over the rationals**, if $V = V' \otimes \mathbb{R}$, where V' is a rational affine space. We say that a polytope $\mathcal{P}$ is **rational** if the extreme points of X are rational.*

Proposition 3.6.2. *Let X be a normal algebraic space, and let V be a finite dimensional affine subspace of the real vector space of Weil divisors which is defined over the rationals.*

Then $\mathcal{L}$ (cf. (1.1.4) for the definition) is a closed rational polytope.

Proof. Note that the set of divisors Δ such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier forms an affine subspace W of V , which is defined over the rationals, so that, replacing V by W , we may assume that $K_X + \Delta$ is $\mathbb{R}$ -Cartier for every $\Delta \in V$.

Let $\pi: Y \rightarrow X$ be a resolution of X , which is a log resolution of the support of any element of V . Given any divisor $\Delta \in V$, we may write

$$K_Y + \Gamma = \pi^*(K_X + \Delta).$$

Then the coefficients of Γ are rational affine linear functions of the coefficients of Δ . On the other hand the condition that $K_X + \Delta$ is log canonical is equivalent to the condition that no component of Γ has coefficient greater than one, and every coefficient of Δ is non-negative. $\square$

Lemma 3.6.3. *Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let $(X, \Delta = A + B)$ be a log pair. Suppose that either*

- (1) *A is big over U and $K_X + \Delta$ is kawamata log terminal, or*
- (2) *A is ample over U and $K_X + \Delta$ is log canonical and there is a divisor Δ'_0 such that $K_X + \Delta'_0$ is kawamata log terminal.*

Let W be the space of Weil divisors on X . Then there is a rational affine linear map

$$L: W \rightarrow W \quad \text{where} \quad \Delta' = A' + B' = L(\Delta),$$

such that

- $K_X + \Delta' \sim_{\mathbb{Q}, U} K_X + \Delta$,
- A' is ample over U ,
- $B' \geq 0$ and
- $K_X + \Delta'$ is kawamata log terminal.

Proof. We may assume that A is a $\mathbb{Q}$ -divisor. Suppose that A is big over U . Then there is a $\mathbb{Q}$ -divisor C , which is ample over U , such that

$$A \sim_{\mathbb{Q}, U} C + C',$$

where $C' \geq 0$. Pick a rational number $\epsilon > 0$ such that

$$K_X + \Delta' = K_X + \epsilon C + (1 - \epsilon)\Delta + \epsilon(B + C') \sim_{\mathbb{Q}, U} K_X + \Delta,$$

is kawamata log terminal and set $A' = \epsilon C$, $B' = (1 - \epsilon)\Delta + \epsilon(B + C')$ and let L be the transformation

$$L(\Theta) = \Theta + \epsilon(C' - A) + A'.$$

Now suppose that A is ample over U . Let V be the vector space spanned by the components of Δ and Δ'_0 . Since $\mathcal{L}$ is a rational polytope, we may find a divisor Δ_0 such that $K_X + \Delta_0$ is kawamata log terminal and $(\Delta - \Delta_0) + 1/2A$ is an ample $\mathbb{Q}$ -divisor over U . But then we may find $A' \sim_{\mathbb{Q},U} (\Delta - \Delta_0) + 1/2A$ such that

$$K_X + \Delta' = K_X + A' + (\Delta_0 - 1/2A) \sim_{\mathbb{Q},U} K_X + \Delta,$$

is kawamata log terminal. Set $B' = \Delta_0 - A/2$ and let L be the transformation

$$L(\Theta) = \Theta + (A' + \Delta_0 - A/2 - \Delta). \quad \square$$

Lemma 3.6.4. *Let $\mathcal{C}$ be a rational polytope contained in a vector space V of dimension n , defined over the rationals. Fix a positive integer k and a positive real number α .*

If $v \in \mathcal{C}$ then we may find a positive integer $m \gg 1$, which is divisible by k , and vectors $v_1, v_2, \dots, v_p \in \mathcal{C}$ such that v is a convex linear combination of the vectors $v_1, v_2, \dots, v_p$ and

$$\|v_i - v\| < \frac{\alpha}{m} \quad \text{where} \quad mv_i \text{ is integral.}$$

Proof. Rescaling by k , we may assume that $k = 1$. We may assume that v is not contained in any proper rational affine linear subspace. In particular, as $\mathcal{C}$ is a rational polytope, it has rational faces and so v is contained in the interior of $\mathcal{C}$.

After fixing a suitable basis for V and possibly shrinking $\mathcal{C}$, we may assume that $\mathcal{C} = [0, 1]^n \subseteq \mathbb{R}^n$ and $v = (x_1, x_2, \dots, x_n) \in (0, 1)^n$. By Diophantine approximation, see [5], there exist $s = (s_1, s_2, \dots, s_n)$ and $b = (b_1, b_2, \dots, b_n) \in (0, 1)^n \cap \mathbb{Q}^n$ and an integer $m \gg 0$, such that, for any $1 \leq i \leq n$, we have that ms_i and mb_i are integers, and

$$s_i \leq x_i \leq b_i, \quad \|v - s\| < \frac{\alpha}{2m} \quad \text{and} \quad \|v - b\| < \frac{\alpha}{2m}.$$

In particular v is contained inside the box

$$\mathcal{B} = [s_1, b_1] \times [s_2, b_2] \times \dots \times [s_n, b_n] \subseteq (0, 1)^n.$$

Then v is a convex linear combination of a subset $v_1, v_2, \dots, v_p$ of the extreme points of $\mathcal{B}$. $\square$

3.7. Rational curves of low degree. We will need the following generalisation of a result of Kawamata, see Theorem 1 of [21], which is proved by Shokurov in the appendix to [32]:

Theorem 3.7.1. *Let $\pi: X \longrightarrow U$ be a projective morphism of normal algebraic spaces. Suppose that (X, Δ) is a log canonical pair of dimension n , where $K_X + \Delta$ is $\mathbb{R}$ -Cartier. Suppose that there is a divisor Δ_0 such that $K_X + \Delta_0$ is kawamata log terminal.*

If R is an extremal ray of $\overline{\text{NE}}(X/U)$ that is $(K_X + \Delta)$ -negative, then there is a rational curve Σ , contracted by π and spanning R , such that

$$0 < -(K_X + \Delta) \cdot \Sigma \leq 2n.$$

Proof. Passing to an étale open subset of U , we may assume that U is affine. Let V be the rational space spanned by the components of $\Delta + \Delta_0$. By (3.6.2) the space $\mathcal{L}$ of log canonical divisors is a rational polytope. Since $\Delta_0 \in \mathcal{L}$, we may find $\mathbb{Q}$ -divisors Δ_i with limit Δ , such that $K_X + \Delta_i$ is kawamata log terminal.

In particular we may assume that $(K_X + \Delta_0) \cdot R < 0$. Replacing π by the contraction defined by the extremal ray R , we may assume that $-(K_X + \Delta)$ is π -ample.

Theorem 1 of [21] implies that we can find a rational curve Σ_i contracted by π such that

$$-(K_X + \Delta_i) \cdot \Sigma_i \leq 2n.$$

Pick a π -ample $\mathbb{Q}$ -divisor A such that $-(K_X + \Delta + A)$ is also π -ample. Now

$$A \cdot \Sigma_i = (K_X + \Delta + A) \cdot \Sigma_i - (K_X + \Delta) \cdot \Sigma_i < 2n.$$

It follows that the curves Σ_i belong to a bounded family. Thus, possibly passing to a subsequence, we may assume that $\Sigma = \Sigma_i$ is constant. In this case

$$-(K_X + \Delta) \cdot \Sigma = \lim_i -(K_X + \Delta_i) \cdot \Sigma \leq 2n. \quad \square$$

Corollary 3.7.2. *Let $\pi: X \longrightarrow U$ be a projective morphism of normal algebraic spaces. Suppose the pair $(X, \Delta = A + B)$ has log canonical singularities, where A is ample over U and $B \geq 0$. Suppose that there is a divisor Δ_0 such that $K_X + \Delta_0$ is kawamata log terminal.*

Then there are only finitely many $(K_X + \Delta)$ -extremal rays $R_1, R_2, \dots, R_k$ of $\overline{\text{NE}}(X/U)$.

Proof. Let R be a $(K_X + \Delta)$ -extremal ray of $\overline{\text{NE}}(X/U)$. Then

$$-(K_X + B) \cdot R = -(K_X + \Delta) \cdot R + A \cdot R > 0.$$

By (3.7.1) R is spanned by a curve Σ such that

$$-(K_X + B) \cdot \Sigma \leq 2n.$$

But then

$$A \cdot \Sigma = -(K_X + B) \cdot \Sigma + (K_X + \Delta) \cdot \Sigma \leq 2n. \quad \square$$

3.8. The directed MMP and the MMP with scaling. To run the MMP with scaling, we will need the following key result:

Lemma 3.8.1. *Let $\pi: X \rightarrow U$ be a projective morphism of normal algebraic spaces. Suppose that the pair $(X, \Delta = A + B)$ has kawamata log terminal singularities, where A is big over U , $B \geq 0$, D is a nef $\mathbb{R}$ -Cartier divisor, but $K_X + \Delta$ is not nef. Set*

$$\lambda = \sup \{ \mu \mid D + \mu(K_X + \Delta) \text{ is nef} \}.$$

Then there is a $(K_X + \Delta)$ -extremal ray R over U , such that

$$(D + \lambda(K_X + \Delta)) \cdot R = 0.$$

Proof. By (3.6.3) we may assume that A is ample over U .

By (3.7.2) there are only finitely many $(K_X + \Delta)$ -extremal rays $R_1, R_2, \dots, R_k$. For each $(K_X + \Delta)$ -extremal ray R_i , pick a curve Σ_i which generates R_i . Let

$$\mu = \min_i \frac{D \cdot \Sigma_i}{-(K_X + \Delta) \cdot \Sigma_i}.$$

Then $D + \mu(K_X + \Delta)$ is nef, since it is non-negative on each R_i , but it is zero on one of the extremal rays $R = R_i$. Thus $\lambda = \mu$. $\square$

Lemma 3.8.2. *Let $\pi: X \rightarrow U$ be a projective morphism of normal algebraic spaces. Suppose the pair $(X, \Delta = A + B)$ has kawamata log terminal singularities, where A is big over U , $B \geq 0$, and C is an $\mathbb{R}$ -Cartier divisor such that $K_X + \Delta$ is not nef, but $K_X + \Delta + C$ is nef and log canonical.*

Then there is a $(K_X + \Delta)$ -extremal ray R and a real number $0 < \lambda \leq 1$ such that $K_X + \Delta + \lambda C$ is nef but trivial on R .

Proof. Apply (3.8.1) to $D = K_X + \Delta + C$. $\square$

Remark 3.8.3. *Assuming existence and termination of the relevant flips, we may use (3.8.2) to define a special minimal model program, which we will refer to as the $(K_X + \Delta)$ -MMP with scaling of C .*

In fact we can also run a slightly more general MMP, which we will call the directed $(K_X + \Delta)$ -MMP. In this MMP, we fix a finite dimensional vector space V of Weil divisors, and an ample divisor A over U . Suppose that after i steps we have reached a model (Y_i, Γ_i) where

Γ_i is the strict transform of Δ , and that we have defined a sequence of divisors $\Delta_1, \Delta_2, \dots, \Delta_i \in \mathcal{L}_A$. At the next step we only consider $(K_{Y_i} + \Gamma_i)$ -extremal rays R , which are $(K_{Y_i} + \Delta'_{i+1})$ -trivial for some Δ'_{i+1} , which is the strict transform of some $\Delta_{i+1} \leq \Delta_i \in \mathcal{L}_A$ on X , and such that $K_{X_i} + \Delta'_{i+1}$ is nef over U .

Note that every MMP with scaling is a directed MMP, and that the only issue with existence of the directed MMP is again termination of flips.

We will need a result about the divisorially log terminal MMP:

Lemma 3.8.4. *Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Suppose that $K_X + \Delta$ is divisorially log terminal and let $\phi: X \dashrightarrow Y$ be a sequence of steps of the $(K_X + \Delta)$ -MMP. Let $\Gamma = \phi_*\Delta$.*

Then

- (1) *ϕ is an isomorphism at the generic point of every log canonical centre of $K_Y + \Gamma$.*
- (2) *If $K_X + \Delta = K_X + S + A + B$, where $S = \lfloor \Delta \rfloor$, A is ample over U and $B \geq 0$, then $K_Y + \Gamma \sim_{\mathbb{R}, U} K_Y + T + C + D$, where $T = \lfloor \Gamma \rfloor$, C is ample over U and $D \geq 0$.*

Proof. We first prove (1). Since the log discrepancy of $K_Y + \Gamma$ is at least the log discrepancy of $K_X + \Delta$ along any valuation ν of X , we may assume that ϕ is either a flip or a divisorial contraction.

Let $p: W \rightarrow X$ and $q: W \rightarrow Y$ be a common log resolution, which resolves the indeterminacy of ϕ . We may write

$$p^*(K_X + \Delta) = q^*(K_Y + \Gamma) + E,$$

E is exceptional and contains every exceptional divisor over the locus where ϕ^{-1} is not an isomorphism. In particular the log discrepancy of every valuation with centre on Y contained in the locus where ϕ^{-1} is not an isomorphism with respect to $K_Y + \Gamma$ is strictly greater than the log discrepancy with respect to $K_X + \Delta$. Hence (1).

Now suppose that $K_X + \Delta = K_X + S + A + B$. Pick a divisor C in Y which is ample over U . Possibly replacing C by a smaller multiple, we may assume that $A - \phi^*C$ is ample. Thus $A - \phi^*C$ is $\mathbb{R}$ -linearly equivalent to a divisor which does not contain any log canonical centre of (X, Δ) . Thus by (1) $\phi_*A - C$ is $\mathbb{R}$ -linearly equivalent to a divisor which does not contain any log canonical centre of (Y, Γ) . (2) is then clear. $\square$

3.9. Shokurov's Polytopes. We will need some results from [35]. First some notation. Given a ray $R \subset \overline{\text{NE}}(X)$, let

$$R^\perp = \{ \Delta \in \mathcal{L} \mid (K_X + \Delta) \cdot R = 0 \}.$$

Theorem 3.9.1. *Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let V be a finite dimensional affine subspace of the real vector space of Weil divisors on X , which is defined over the rationals. Fix A , an ample divisor over U . Suppose that there is a divisor Δ_0 such that $K_X + \Delta_0$ is kawamata log terminal.*

Then the set of hyperplanes $R^\perp$ is finite in $\mathcal{L}_A$, as R ranges over the set of extremal rays of $\overline{\text{NE}}(X/U)$. In particular $\mathcal{N}_A$ is a closed rational polytope.

Corollary 3.9.2. *Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties. Let V be a finite dimensional affine subspace of the real vector space of Weil divisors on X , which is defined over the rationals. Fix A , an ample $\mathbb{Q}$ -Cartier divisor over U . Let $\phi: X \dashrightarrow Y$ be any birational map over U , whose inverse does not contract any divisors. Suppose that there is a divisor Δ_0 such that $K_X + \Delta_0$ is kawamata log terminal.*

Then $\mathcal{P}_Y$ is a rational polytope.

Proof of (3.9.1). Since $\mathcal{L}_A$ is compact it suffices to prove this locally about any point $\Delta \in \mathcal{L}_A$. By (3.6.3) we may assume that $K_X + \Delta$ is kawamata log terminal. Fix $\epsilon > 0$ such that if $\|\Delta' - \Delta\| < \epsilon$, then $\Delta' - \Delta + A/2$ is ample. Let R be an extremal ray such that $(K_X + \Delta') \cdot R = 0$, where $\|\Delta' - \Delta\| < \epsilon$. We have

$$(K_X + \Delta - A/2) \cdot R = (K_X + \Delta') \cdot R - (\Delta' - \Delta + A/2) \cdot R < 0.$$

Finiteness then follows from (3.7.2).

$\mathcal{N}_A$ is surely a closed subset of $\mathcal{L}_A$. If $K_X + \Delta$ is not nef then (3.7.1) implies that $K_X + \Delta$ is negative on a rational curve Σ which generates an extremal ray R of $\overline{\text{NE}}(X/U)$. Thus $\mathcal{N}_A$ is the intersection of $\mathcal{L}_A$ with the finitely many half-spaces determined by the extremal rays of $\overline{\text{NE}}(X/U)$. $\square$

Proof of (3.9.2). We may as well suppose that V is the space spanned by the components of a divisor D . By compactness, it is enough to show that $\mathcal{P}_Y$ is locally a rational polytope.

Let $\Delta \in \mathcal{P}_Y$. By (2) of (3.6.3) we may assume that $K_X + \Delta$ is kawamata log terminal, so that $K_Y + \Gamma$ is kawamata log terminal as well. Let $C = \phi_* A$. Then C is big over U . By (1) of (3.6.3) and (3.9.1) we may assume that $\mathcal{N}_C$ is locally a rational polytope about Γ .

Let $p: W \longrightarrow X$ be a log resolution of (X, Δ) which resolves the indeterminacy locus of ϕ , via a birational map $q: W \longrightarrow Y$. We may write

$$\begin{aligned} K_W + \Psi &= p^*(K_X + \Delta) \\ K_W + \Phi &= q^*(K_Y + \Gamma). \end{aligned}$$

Note that $\Delta \in \mathcal{P}_Y$ iff $\Gamma = \phi_*\Delta \in \mathcal{N}_C$ and $\Psi - \Phi \geq 0$. Since the assignment $\Delta \longrightarrow \Gamma$ is rational linear, the result is clear. $\square$

4. SPECIAL TERMINATION WITH SCALING

In this section we show that we have special termination of the MMP with scaling.

Lemma 4.1. *Assume Theorem B_n and Theorem C_n .*

Let $\pi_i: X_i \longrightarrow U$ be a sequence of projective morphisms of normal quasi-projective varieties, where X_i and X_j are isomorphic in codimension one over U . Suppose that $K_{X_i} + \Delta_i$ is kawamata log terminal and nef over U . Suppose that there are fixed divisors A_1 and B_1 on $X = X_1$, with transforms A_i and B_i on X_i , such that

$$A_i \leq \Delta_i \leq B_i,$$

where A_1 is big over U and $K_{X_1} + B_1$ is kawamata log terminal. Let n be the dimension of X .

Then the set of isomorphism classes

$$\{X_i \mid i \in \mathbb{N}\}$$

is finite.

Proof. Let Δ'_i be the strict transform of Δ_i on X . By (1) of (3.6.3) we may assume that A_1 is ample over U . As (X_i, Δ_i) is a log terminal model of (X, Δ'_i) , the result follows as we are assuming Theorem B_n and Theorem C_n . $\square$

Lemma 4.2. *Assume Theorem B_{n-1} and Theorem C_{n-1} .*

Let $\pi: X \longrightarrow U$ be a projective morphism of normal quasi-projective varieties, where X is a $\mathbb{Q}$ -factorial variety of dimension n . Suppose that

$$K_X + \Delta = K_X + S + A + B,$$

is divisorially log terminal and nef over U , where A is ample over U and $B \geq 0$.

Then every sequence of flips over U for the directed $(K_X + \Delta)$ -MMP is eventually disjoint from $S = \lfloor \Delta \rfloor$.

Proof. Suppose not. Let $X_i \dashrightarrow X_{i+1}$ be an infinite sequence of flips over U , starting with $X_1 := X$, for the directed $(K_X + \Delta)$ -MMP, which meets S infinitely often. Then there is a component T of S that intersects the flipping locus infinitely many times. Pick a rational number $\epsilon > 0$ such that $A' = A + \epsilon(S - T)$ is ample over U . Replacing A' by an $\mathbb{R}$ -linearly equivalent over U divisor, we may assume that

$$K_X + S - \epsilon(S - T) + A' + B \sim_{\mathbb{R}, U} K_X + S + A + B,$$

is purely log terminal. Note that every step of the $(K_X + S + A + B)$ -MMP over U is a step of the $(K_X + S - \epsilon(S - T) + A' + B)$ -MMP over U , and $\lfloor S - \epsilon(S - T) + A' + B \rfloor = T$. Thus, replacing $K_X + \Delta$ by $K_X + S - \epsilon(S - T) + A' + B$ we may assume that S is irreducible and $K_X + \Delta$ is purely log terminal. Let S_i be the strict transform of S in X_i and let $S_i \dashrightarrow S_{i+1}$ be the induced birational map.

By [9], we may assume that $X_i \dashrightarrow X_{i+1}$ is an isomorphism at the generic point of every divisor on S_i and S_{i+1} . In particular we may assume that $S_i \dashrightarrow S_{i+1}$ is an isomorphism in codimension one.

Let Δ_i be the strict transform of Δ on X_i . By adjunction we may write

$$(K_{X_i} + \Delta_i)|_{S_i} = K_{S_i} + \Theta_i.$$

Then Θ_i is the strict transform of Θ_1 . By definition of a directed MMP, the hypotheses of (4.1) are satisfied, and the set of isomorphism classes

$$\{S_i \mid i \in \mathbb{N}\},$$

is finite, so that the set of pairs

$$\{(S_i, \Theta_i) \mid i \in \mathbb{N}\},$$

is also finite.

On the other hand, let $X_i \rightarrow Z_i$ be the flipping contraction and let T_i be the normalisation of the image of S_i in Z_i , so that there are birational morphisms $p_i: S_i \rightarrow T_i$ and $q_i: S_{i+1} \rightarrow T_i$. Note that $-(K_{S_i} + \Theta_i)$ is p_i -ample whilst $(K_{S_{i+1}} + \Theta_{i+1})$ is q_i -ample. By assumption infinitely often the flipping locus intersects S_i . If p_i is an isomorphism and the flipping locus intersects S_i , then $S_i \cdot \Sigma_i > 0$, where Σ_i is a flipping curve. But then S_{i+1} must intersect the flipped curve negatively, so that the flipped curve lies in S_{i+1} and q_{i+1} is not an isomorphism. In particular infinitely often one of the birational morphisms p_i or q_i is not an isomorphism.

Thus we may assume that p_1 or q_1 is not an isomorphism, where the isomorphism class of S_1 is repeated infinitely often. Pick any valuation ν whose centre is contained in the locus where $S_1 \dashrightarrow S_2$ is not an

isomorphism. By (2.28) of [25]

$$a(\nu, S_1, \Theta_1) < a(\nu, S_2, \Theta_2) \quad \text{and} \quad a(\nu, S_i, \Theta_i) \leq a(\nu, S_{i+1}, \Theta_{i+1}),$$

a contradiction. $\square$

We use (4.2) to run a special MMP:

Lemma 4.3. *Assume Theorem B_{n-1} and Theorem C_{n-1} .*

Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties, where X is $\mathbb{Q}$ -factorial of dimension n . Suppose that $(X, \Delta + C = S + A + B + C)$ is a divisorially log terminal pair, such that $\lfloor \Delta + C \rfloor = S$, A is ample over U , and $B, C \geq 0$. Suppose that there is an $\mathbb{R}$ -divisor $D \geq 0$ whose support is contained in S and a real number $\alpha \geq 0$, such that

$$(*) \quad K_X + \Delta \sim_{\mathbb{R}, U} D + \alpha C.$$

If $K_X + \Delta + C$ is nef over U then there is a log terminal model $\phi: X \dashrightarrow Y$ for $K_X + \Delta$ over U .

Proof. Since X is $\mathbb{Q}$ -factorial, (2) of (3.6.3) implies that we may find $\Delta' \sim_{\mathbb{R}, U} \Delta$ such that $K_X + \Delta'$ is kawamata log terminal. In particular we may apply (3.8.2) to $K_X + \Delta$ to run the MMP with scaling over U .

We run the the $(K_X + \Delta)$ -MMP over U with scaling of C . Pick $t \in [0, 1]$ minimal such that $K_X + \Delta + tC$ is nef over U . If $t = 0$ we are done. Otherwise we may find a $(K_X + \Delta)$ -negative extremal ray R over U , such that $(K_X + \Delta + tC) \cdot R = 0$. Let $f: X \rightarrow Z$ be the associated contraction over U . As $t > 0$, $C \cdot R > 0$ and so $D \cdot R < 0$. In particular f is always birational.

If f is divisorial, then we can replace X, S, A, B, C and D by their images in Z . Note that $(*)$ continues to hold.

Otherwise f is small. As $D \cdot R < 0$, R is spanned by a curve Σ which is contained in a component T of S , where $T \cdot \Sigma < 0$. Note that $K_X + S + A + B - \epsilon(S - T)$ is a purely log terminal divisor for any positive $\epsilon \ll 1$, and so f is a pl flip.

As we are assuming Theorem B_{n-1} and Theorem C_{n-1} , (1.4.2) $_{n-1}$ holds and so the main result of [13] implies that the flip $f': X' \rightarrow Z$ of $f: X \rightarrow Z$ exists. Again, if we replace X, S, A, B, C and D by their images in X' , then $(*)$ continues to hold. On the other hand this flip is certainly not an isomorphism in a neighbourhood of S and so the MMP terminates by (4.2). Note that even though the strict transform of A is not ample, in view of (3.8.4) we can replace Δ by an $\mathbb{R}$ -linearly equivalent divisor, such that the hypothesis of (4.2) are satisfied $\square$

5. LOG TERMINAL MODELS

We need an ad hoc definition of the stable base locus of a real divisor:

Definition 5.1. *Let D be an $\mathbb{R}$ -divisor on a locally Noetherian scheme. The **real linear system** associated to D is*

$$|D|_{\mathbb{R}} = \{ C \geq 0 \mid C \sim_{\mathbb{R}} D \}.$$

*The **real stable base locus** of D is the intersection of the elements of the real linear system $|D|_{\mathbb{R}}$. The **real stable fixed divisor** is the divisorial support of the real stable base locus.*

Remark 5.2. *The real stable base locus and the real stable fixed divisor are only defined as closed subsets, they do not have any scheme structure.*

Lemma 5.3. *Let $\pi: X \rightarrow U$ be a birational morphism of normal projective varieties, let $D \geq 0$ be an $\mathbb{R}$ -Cartier divisor on U and let E be an effective $\mathbb{R}$ -divisor on X such that π_*E is supported on the real stable fixed divisor of D .*

*If $G \in |\pi^*D + E|_{\mathbb{R}}$ then $G - E \geq 0$. In particular the real stable fixed divisor of π^*D is contained in the real stable fixed divisor of $\pi^*D + E$.*

Proof. Cancelling we may assume that no component of E is a component of G , and it suffices to prove that $E = 0$. Suppose not.

By induction we may assume that E is irreducible. Rescaling we may assume that E is a prime divisor. If E is exceptional then E is covered by a family of curves Σ such that $E \cdot \Sigma < 0$. But then $G \cdot \Sigma < 0$ so that $\text{mult}_E G > 0$, a contradiction.

Otherwise E is in the real stable fixed divisor of π^*D . Hence we may assume that π is the identity. Let $k = \text{mult}_E D$. Then $k > 0$ as $D \geq 0$ and E is a component of the stable fixed divisor. Set

$$C = D + k(G - E).$$

By construction $\text{mult}_E C \geq 0$, so that $C \geq 0$. Now

$$C \sim_{\mathbb{R}} (k + 1)D,$$

so that $\text{mult}_E C > 0$ by definition of the real stable fixed divisor. But then $\text{mult}_E G > 0$, a contradiction. $\square$

Lemma 5.4. *Let $\pi: X \rightarrow U$ be a morphism of normal projective varieties. Suppose that (X, Δ) is a kawamata log terminal pair such that Δ is big over U and $K_X + \Delta \sim_{\mathbb{R}, U} D \geq 0$, for some $\mathbb{R}$ -divisor D . Suppose that $K_X + \Theta$ is kawamata log terminal and that $\Theta - \Delta$ is supported on the real stable fixed divisor of $K_X + \Delta$. Let $\phi: X \dashrightarrow Y$ be a birational map over U , whose inverse does not contract any divisors.*

Suppose that Y is $\mathbb{Q}$ -factorial and that $K_Y + \Gamma$ is nef and kawamata log terminal, where $\Gamma = \phi_*\Theta$.

If ϕ only contracts components of the real stable fixed divisor of $K_X + \Delta$, then ϕ is a log terminal model of (X, Δ) over U .

Proof. We first show that ϕ is a log terminal model of $K_X + \Theta$. Let $p: W \rightarrow X$ and $q: W \rightarrow Y$ resolve the indeterminacy of ϕ . We may write

$$p^*(K_X + \Theta) + E = q^*(K_Y + \Gamma) + F,$$

where E and F are effective, with no common components, and both E and F are exceptional for q , and the support of F is the whole exceptional locus.

Let $G = p_*E$. Then G is supported on the real stable fixed divisor of $K_X + \Theta$. (5.3) implies that the support of $p^*(K_X + \Theta) + E$ contains every q -exceptional divisor which is not p -exceptional.

Negativity of contraction implies that it suffices to prove that $G = 0$. Suppose not. Pick a component B of E which is not contracted by p . As B is a component of E it cannot be a component of F . By the base point free theorem, cf. (3.4.1), $q^*(K_Y + \Gamma)$ is semiample over U . But this contradicts the fact that B is a component of the real stable fixed divisor of $p^*(K_X + \Theta) + E$. Thus ϕ is a log terminal model of $K_X + \Theta$.

By (5.3) the support of the real stable fixed divisor of $K_X + \Delta$ and $K_X + \Theta$ are the same. Thus ϕ must contract every component of the real stable fixed divisor of Δ . In particular $\phi_*\Delta = \phi_*\Theta = \Gamma$. But then by what we have already proved, ϕ is a log terminal model of (X, Δ) over U . $\square$

Lemma 5.5. *Suppose that*

$$0 < m_1 < m_2 < m_3 < \cdots < m_{l+1},$$

is an increasing sequence of positive real numbers.

Then, for $1 \leq i \leq l$ and $1 \leq j \leq l+1$, the system of equations

$$\begin{aligned} \text{(i)} \quad & \alpha_i n_{ij} - (\alpha_i + 1) n_{i+1,j} = m_j \quad \text{for } 1 \leq j \leq i+1 \leq l \\ \text{(ii)} \quad & n_{ij} = 1 \quad \text{for } i < j \leq l+1 \\ \text{(iii)} \quad & n_{lj} = \frac{m_j}{m_{l+1}} \quad \text{for } j \leq l+1, \end{aligned}$$

for the real numbers $\alpha_1, \alpha_2, \dots, \alpha_{l-1}$ and the $l \times (l+1)$ real matrix (n_{ij}) has a unique solution. Further this solution satisfies

- (1) $0 < \alpha_i$, for all $1 \leq i \leq l-1$,
- (2) $0 < n_{i+1,j} < n_{ij}$, for all $1 \leq i < l-1$,
- (3) $0 < n_{ij}$, for all $1 \leq i \leq l, 1 \leq j \leq l+1$,

- (4) $n_{ij} < n_{i,j+1}$, for all $1 \leq j \leq i \leq l$, and
- (5) $n_{ij} < 1$, for all $1 \leq j \leq i \leq l$.

Proof. We proceed by decreasing induction on i to determine the values of α_i and n_{ij} . When $i = l$, (iii) determines the value of n_{lj} . Further these values are consistent with (ii), for $i = l$ and $j = l + 1$ and (3), (4) and (5) hold.

Suppose that we have determined the values of $n_{i'j}$ for $i + 1 \leq i' \leq l$ and $\alpha_{i'}$ for $i + 1 \leq i' < l$. (ii) determines the values of $n_{ij} = 1$, for $j > i$. Note that $n_{i+1,i+1} < 1 = n_{i,i+1}$ so (2) holds in this case. Consider the equation from (i)

$$\alpha_i n_{i,i+1} - (\alpha_i + 1) n_{i+1,i+1} = m_{i+1}.$$

Since we know that $n_{i,i+1} = 1$ and we have already determined the value of $n_{i+1,i+1}$, we can use this equation to determine the value of

$$\alpha_i = \frac{n_{i+1,i+1} + m_{i+1}}{1 - n_{i+1,i+1}} > 0,$$

which is (1). Finally we then use this value of α_i , the values of $n_{i+1,i}$, and the remaining equations from (i)

$$\alpha_i n_{ij} - (\alpha_i + 1) n_{i+1,j} = m_j \quad \text{for } 1 \leq j \leq i \leq l - 1,$$

to determine the value of

$$n_{ij} = \frac{(\alpha_i + 1) n_{i+1,j} + m_j}{\alpha_i}.$$

(2) is clear and this implies (3). (4) is also clear and this implies (5). $\square$

Lemma 5.6. *Assume Theorem B_{n-1} and Theorem C_{n-1} .*

Let $\pi: X \rightarrow U$ be a morphism of normal projective varieties, where X has dimension n . Let D and $\Delta = A + B$ be two $\mathbb{R}$ -divisors with the following properties:

- (1) *the support of D and Δ has global normal crossings (in particular X is smooth),*
- (2) $\lrcorner \Delta \lrcorner = 0$,
- (3) *A is ample over U and $B \geq 0$,*
- (4) $K_X + \Delta \sim_{\mathbb{R},U} D \geq 0$,
- (5) *$D = M + F$, where every component of M is mobile and every component of F is a component of the real stable fixed divisor of D , and*
- (6) *Δ and M have no components in common.*

Then there is a log terminal model $\phi: X \dashrightarrow Y$ for (X, Δ) over U .

Proof. Let Θ_0 be the divisor which contains the support of $M + F$ with coefficient one and otherwise equals Δ . Let k be the number of distinct coefficients in M . The idea is to choose a sequence of divisors

$$\Delta \leq \Theta_k \leq \Theta_{k-1} \leq \cdots \leq \Theta_0,$$

and to successively construct a log terminal model for each $K_X + \Theta_i$ over U , so that we are going to apply (4.3) $k + 1$ times. At the i th step, denote by N_i the part of Θ_i with support equal to the support of M (to simplify the notation, we use indices only on those divisors whose coefficients vary from step to step; if the coefficients do not change, we will abuse notation and use the same symbol for a divisor and its strict transform). We can break the $k + 1$ applications of (4.3) into three parts:

Start (Easy: zeroth step) Reduce to the case $K_X + \Theta_0$ is nef over U .
Middle (Hard: next $k - 1$ steps) Reduce to the case where N_{k-1} is a scalar multiple of M .

End (Medium: last step) Eliminate M .

At each step, the support of $\Psi_i = \Theta_i - \Delta \geq 0$ is contained in the support of D . Suppose that E is a divisor which is contracted at the i th step. Then we will prove that E must be a component of D . On the other hand, as every component of M is mobile, no component of M is ever contracted. Thus E is a component of the real stable fixed divisor. By (5.4) we are therefore done as soon as $K_Y + \Gamma$ is nef over U , where $\Gamma = \phi_*\Delta$.

Note that $K_X + \Theta_0$ is divisorially log terminal. Pick an ample divisor H over U , such that $K_X + \Theta_0 + H$ is ample over U . Let $S_0 = \lrcorner \Theta_0 \lrcorner$, $B_0 = \Theta_0 - S_0 - A$ and $C = H$, so that

$$K_X + \Theta_0 + H = K_X + S_0 + A + B_0 + C.$$

If we put $D_0 = D + \Theta_0 - \Delta \sim_{\mathbb{R}} K_X + \Theta_0$ then the support of D_0 is contained in S_0 . By (4.3) we may assume that $K_X + \Theta_0$ is nef over U .

Suppose that

$$M = \sum_{i=1}^k m_i M_i \quad \text{so that} \quad N_0 = \sum_{i=1}^k M_i.$$

Possibly reordering and grouping together divisors with the same coefficient, we may assume that

$$0 < m_1 < m_2 < m_3 < \cdots < m_k.$$

For every $1 \leq i \leq k-1$, we may write,

$$N_i = \sum_{j=1}^k n_{ij} M_j,$$

where the coefficients n_{ij} are given by (5.5). For any $1 \leq i \leq k-1$, choose Θ_i so that

$$\Psi_i = \Theta_i - \Delta = N_i + G \geq 0,$$

where $\text{Supp } N_i = \text{Supp } M$ and $\text{Supp } G = \text{Supp } F$.

Then

$$S_i = \lfloor \Theta_i \rfloor = M_{i+1} + M_{i+2} + \cdots + M_k + \lceil G \rceil.$$

Suppose that, after i steps, we have constructed a log terminal model for (X, Θ_i) over U , for $i \geq 0$, so that $K_X + \Theta_i$ is nef over U . By (3.8.4), possibly replacing Θ_i by an $\mathbb{R}$ -linearly equivalent divisor, we may assume that Θ_i contains an ample divisor over U . We will abuse notation and continue to refer to this divisor as A .

Let $C_i = \Theta_i - \Theta_{i+1}$. Property (2) of (5.5) ensures that $C_i \geq 0$. Define $B_{i+1} = \Theta_i - S_{i+1} - A - C_i$. If $K_X + \Theta_i$ is nef over U then, at the next step,

$$K_X + \Theta_{i+1} = K_X + S_{i+1} + A + B_{i+1},$$

is nef over U .

Since

$$n_{k-1,j} = \frac{m_j}{m_k},$$

after running the $k-1$ minimal model programs we end up with

$$N_{k-1} = \lambda M \quad \text{where} \quad \lambda = \frac{1}{m_k}.$$

Finally we must choose an $\mathbb{R}$ -divisor $D_{i+1} \geq 0$ and a constant $\alpha_i > 0$ so that

$$K_X + \Theta_{i+1} \sim_{\mathbb{R}} D_{i+1} + \alpha_i C_i,$$

and the support of D_{i+1} is contained in S_{i+1} .

Since $\Theta_i = \Theta_{i+1} + C_i$, it is enough to choose D_{i+1} and α_i so that

$$D_{i+1} + (\alpha_i + 1)C_i \sim_{\mathbb{R}} K_X + \Theta_i = M + N_i + F + G,$$

and the coefficient of M_j in D_{i+1} is zero for $j \leq i+1$, so that the support of D_{i+1} is contained in S_{i+1} . Since the coefficient of M_j in C_i is zero for $j > i+1$, it automatically follows that $D_{i+1} \geq 0$. Substituting $C_i = N_i - N_{i+1}$ and equating the coefficients of N_j for $j \leq i+1$, we get

$$\alpha_i n_{ij} - (\alpha_i + 1) n_{i+1,j} = m_j,$$

after a little simplification. By (5.5) we may indeed solve for α_i and n_{ij} .

Applying (4.3) $k - 1$ times, we may assume that $K_X + \Theta_{k-1}$ is nef over U , where

$$\Psi_{k-1} = \Theta_{k-1} - \Delta = N_{k-1} + G = \lambda M + G.$$

Let

$$S_k = \lceil G \rceil = \lfloor \Theta_{k-1} \rfloor - M_k,$$

$C_k = N_{k-1} = \lambda M$ and $B_k = \Theta_{k-1} - S_k - C_k - A$. Then

$$K_X + \Theta_k = K_X + S_k + A + B_k \sim_{\mathbb{R}} F + G + \alpha_k C_k,$$

where $\alpha_k = \frac{1}{\lambda}$. Thus by (4.3) we may assume that

$$K_X + \Theta_k = K_X + \Delta + G,$$

is nef over U , and we are done by (5.4). $\square$

Lemma 5.7. *Let X be a normal projective variety and let B be a prime divisor that is not contained in the real stable fixed divisor of the $\mathbb{R}$ -divisor $D \geq 0$.*

Then there are infinitely many integers m , such that B is not a component of the base locus of the linear system of $\lfloor mD \rfloor$.

Proof. We may assume that B is a component of D , otherwise there is nothing to prove. As B is not contained in the real stable fixed divisor of D , we may find $D' \in |D|_{\mathbb{R}}$ such that B is not a component of D' . As $D \sim_{\mathbb{R}} D'$, there are positive real numbers r_i and rational functions f_i , such that

$$D = D' + R \quad \text{where} \quad R = \sum r_i(f_i) \sim_{\mathbb{R}} 0.$$

By Diophantine approximation, see [5], we may find an integer $m \gg 1$ and an integral divisor

$$R_m = \sum m_i(f_i) \sim 0,$$

such that if

$$E = mR - R_m = \sum e_i(f_i),$$

then $0 < e_i < 1$ and

$$\text{mult}_B E = \sum e_i \text{mult}_B(f_i) < 1.$$

Thus,

$$\begin{aligned} \lfloor mD \rfloor &= \lfloor mD' + mR \rfloor \\ &= \lfloor mD' + E \rfloor + R_m \\ &\sim C, \end{aligned}$$

where $C = \lfloor mD' + E \rfloor$. Since $D \geq 0$, every divisor with negative coefficient in R , must occur in D' with strictly positive coefficient. As the only components of E which occur with negative coefficient are poles of f_i , for some i , it follows that $C \geq 0$ for m sufficiently large. On the other hand, as B is not a component of D' and $\text{mult}_B E < 1$, it follows that B is not a component of C . Thus the base locus of the linear system of $\lfloor mD \rfloor$ does not contain B in its support. $\square$

Lemma 5.8. *Let X be a normal projective variety and let $D \geq 0$ be an $\mathbb{R}$ -divisor. Then we may find $\mathbb{R}$ -divisors M and F such that*

- (1) $M \geq 0$ and $F \geq 0$,
- (2) $D \sim_{\mathbb{R}} M + F$,
- (3) *every component of F is a component of the real stable fixed divisor, and*
- (4) *every component of M is mobile.*

Proof. We may write $D = M + F$ where every component of F is a component of the real stable fixed locus and no component of M is a component of the real stable fixed divisor of D . We proceed by induction on the number of components of M which are not mobile. Suppose that B is a component of M which is not mobile. Rescaling we may assume that the coefficient of B in D is one. As B is not a component of the real stable fixed divisor of D , (5.7) implies that there exists an integer $m \gg 1$ such that the base locus of the linear system of $\lfloor mD \rfloor$ does not contain B in its support. Let C be a general element of the linear system $|\lfloor mD \rfloor|$. Then C is a sum of mobile divisors and components of D other than B . In this case

$$\begin{aligned} D &= \frac{1}{m} \lfloor mD \rfloor + \frac{1}{m} \{mD\} \\ &\sim_{\mathbb{R}} \frac{1}{m} C + \frac{1}{m} \{mD\}. \end{aligned}$$

Note that $\{mD\}$ does not contain B in its support as $\text{mult}_B D = 1$. Thus replacing D by $C/m + \{mD\}/m$ we are done by induction. $\square$

Lemma 5.9. *Theorem B_{n-1} and Theorem C_{n-1} imply Theorem A_n .*

Proof. Let U_0 be a compactification of U , and let X_0 be a compactification of X , lying over U_0 . Let Δ_0 be the closure of Δ . By (3.5.5) we may assume that (X_0, Δ_0) is log smooth. By (3.2.1) we may assume that $K_{X_0} + \Delta_0 \sim_{\mathbb{R}, U_0} D_0 \geq 0$. Since a log terminal model over U_0 restricts to a log terminal model over U , replacing $X \rightarrow U$ by $X_0 \rightarrow U_0$, we may assume that U is projective.

Pick an ample Cartier divisor H on U such that $K_X + \Delta + \pi^*H \sim_{\mathbb{R}} D + \pi^*H \geq 0$. Replacing Δ by $\Delta + \pi^*H$ and D by $D + \pi^*H$, we may

assume that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$. By (1) of (3.6.3) we may assume that $\Delta = A + B$, where A is ample over U and B is effective. By (5.8) we may assume that $D = M + F$, where every component of F is a component of the real stable fixed divisor and every component of M is mobile. We may assume that every component B of M is general in $|B|$. Pick a log resolution $f: Y \rightarrow X$ of the support of D and Δ , which resolves the base locus of the components of M . Let Γ be the divisor defined in (3.5.5). Note that every component $\tilde{B}$ of the strict transform $\tilde{M}$ of M is still mobile and every component of the exceptional locus belongs to the real stable fixed divisor. Possibly replacing Γ by an $\mathbb{R}$ -linearly equivalent divisor, we may assume that Γ contains an ample divisor. By (3.5.5), we may replace X by Y and the result follows by (5.6). $\square$

6. FINITENESS OF MODELS

Lemma 6.1. *Assume Theorem A_n . Let $\pi: X \rightarrow U$ be a projective morphism of normal quasi-projective varieties, where X has dimension n . Fix A , an ample divisor over U . Suppose that there is a divisor Δ'_0 such that $K_X + \Delta'_0$ is kawamata log terminal. Let $\mathcal{C} \subset \mathcal{L}_A$ be a subset with the following properties:*

- (1) *For every $\Delta \in \mathcal{C} \cap \mathcal{P}_A$, there is a divisor $D \geq 0$ with $K_X + \Delta \sim_{\mathbb{R}, U} D$.*
- (2) *$\mathcal{C}$ is a polytope.*

Then we can find rational maps $\phi_i: X \dashrightarrow Y_i$ over U , $1 \leq i \leq k$, such that if $\Delta \in \mathcal{C} \cap \mathcal{P}_A$ then ϕ_i is a log terminal model of $K_X + \Delta'$ over U , for some i , for any $K_X + \Delta' \sim_{\mathbb{R}} K_X + \Delta$ which is kawamata log terminal. In particular the set of isomorphism classes

$$\{Y \mid Y \text{ is the log canonical model of a pair } (X, \Delta), \text{ where } \Delta \in \mathcal{C}\},$$

is finite.

Proof. By compactness it suffices to prove this result locally about any divisor $\Delta_0 \in \mathcal{C}$. By (2) of (3.6.3), we may assume that $K_X + \Delta_0$ is kawamata log terminal.

If $K_X + \Delta_0$ is not π -pseudo-effective then there is an open neighbourhood P_0 of Δ_0 in $\mathcal{L}_A$, such that $K_X + \Delta$ is not π -pseudo-effective for all $\Delta \in P_0$. In this case there are no log terminal models in a neighbourhood of Δ_0 .

Thus we may assume that $\Delta_0 \in \mathcal{P}_A$. As we are assuming Theorem A_n there is a log terminal model $\phi: X \dashrightarrow Y$ over U for $K_X + \Delta_0$. In particular ϕ is $(K_X + \Delta_0)$ -negative. But then there is an open

neighbourhood P_0 of Δ_0 in $\mathcal{C}$ such that ϕ is $(K_X + \Delta)$ -negative for any $\Delta \in P_0$.

Thus replacing X by Y we may assume that $K_X + \Delta_0$ is π -nef. Let $X \rightarrow Z$ be the ample model of $K_X + \Delta_0$. In particular, $K_X + \Delta_0 \sim_{\mathbb{R}, Z} 0$. Since the boundary of $\mathcal{C}$ lies in finitely many rational affine hyperplanes, we can assume, by induction on the dimension of V , that there are finitely many birational maps $\phi_i: X \dashrightarrow Y_i$ over Z , $1 \leq i \leq k$, such that for any Δ on the boundary of $\mathcal{C}$, where $K_X + \Delta$ is pseudo-effective over Z , there is an index i , such that ϕ_i is a log terminal model of $K_X + \Delta$ over Z .

Let $\Gamma_0 = \phi_{i*}\Delta_0$ and $\Gamma = \phi_{i*}\Delta$. Note that $K_{Y_i} + \Gamma_0$ is numerically trivial over Z and nef over U . By (3.9.1), for each $1 \leq i \leq k$, there is a positive constant ϵ_i , depending on i but not on Δ , such that if R is an extremal ray of $\overline{\text{NE}}(Y_i/U)$, which is $(K_{Y_i} + (1 - \epsilon_i)\Gamma_0 + \epsilon_i\Gamma)$ -negative, then R is also an extremal ray of $\overline{\text{NE}}(Y_i/Z)$. As $K_{Y_i} + (1 - \epsilon_i)\Gamma_0 + \epsilon_i\Gamma$ is nef over Z , it is nef over U .

Now let $\epsilon := \min_{1 \leq i \leq k} \{\epsilon_i\}$. Pick $\Theta \in \mathcal{C}$ such that $\|\Theta - \Delta_0\| < \epsilon$. Pick Δ on the boundary of $\mathcal{C}$ such that $\Theta \in (\Delta_0, \Delta]$. Suppose that $K_X + \Theta$ is π -pseudo-effective. Then

$$\Theta - \Delta_0 = (K_X + \Theta) - (K_X + \Delta_0),$$

is pseudo-effective over Z . But then $K_X + \Delta$ is also pseudo-effective over Z . In this case there is an index i such that Y_i is a log terminal model for

$$K_X + (1 - \lambda)\Delta_0 + \lambda\Delta,$$

over U , for any $\lambda \in (0, \epsilon]$. In particular Y_i is a log terminal model for $K_X + \Theta$ over U . $\square$

We will need a variation on (3.5.5):

Lemma 6.2. *Let $\pi: X \rightarrow U$ be a projective morphism of normal algebraic spaces. Let (X, Δ) be a kawamata log terminal pair, and let $\xi: X \dashrightarrow Y$ be a weak log canonical model of $K_X + \Delta$ over U . Let $\Gamma = \xi_*\Delta$. Let $f: Z \rightarrow X$ be any log resolution of (X, Δ) and suppose that we write*

$$K_Z + \Phi_0 = f^*(K_X + \Delta) + E,$$

where Φ_0 and E are effective, with no common components, $f_\Phi_0 = \Delta$ and E is exceptional. Let F be any divisor whose support is equal to the exceptional locus.*

Then we may find $\eta > 0$ such that if $\Phi = \Phi_0 + \eta F$ then

- $f_*\Phi = \Delta$,
- $K_Z + \Phi$ is kawamata log terminal,

- if Δ is big over U then so is Φ , and
- every log terminal model (respectively weak log canonical model) over Y of $K_Z + \Phi$ is a log terminal model (respectively weak log canonical model) over U of $K_X + \Delta$.

Proof. Everything is clear, apart from the fact that if $\phi: Z \dashrightarrow W$ is a log terminal model (respectively weak log canonical model) over Y of $K_Z + \Phi$ then it is a log terminal model (respectively weak log canonical model) over U of $K_X + \Delta$.

Let $\psi: X \dashrightarrow W$ be the induced birational map and set $\Psi = \phi_*\Phi$. By assumption there is a morphism $g: W \rightarrow Y$. By (3.5.5) possibly blowing up more, we may assume that ϕ is a morphism. By assumption if we write

$$K_Z + \Phi = \phi^*(K_W + \Psi) + G,$$

then $G \geq 0$ and the support of G is the full ϕ -exceptional locus (respectively G is exceptional). Similarly we may write

$$f^*(K_X + \Delta) = \phi^*(g^*(K_Y + \Gamma)) + E',$$

where $E' \geq 0$ is exceptional over Y . Thus

$$\phi^*(K_W + \Psi) + G = \phi^*(g^*(K_Y + \Gamma)) + E' + E + \eta F.$$

As every component of $G + E' + E + \eta F$ is exceptional over Y , by negativity of contraction $G - E' - E - \eta F \geq 0$. In particular every f -exceptional divisor is a component of G , so that ϕ contracts every f -exceptional divisor and so ψ^{-1} does not contract any divisors. Then $g_*\Psi = \Gamma$. As $K_W + \Psi$ is nef over Y , it follows that $(K_W + \Psi) = g^*(K_Y + \Gamma)$. But then ϕ is a log terminal model (respectively weak log canonical model) over U by (3.5.3). $\square$

Lemma 6.3. *Theorem B_{n-1} and Theorem A_n imply Theorem B_n .*

Proof. By (3.5.5) we may assume that X is smooth. Since the property of a birational map $\phi: X \dashrightarrow Y$ being a weak log canonical model of a log canonical pair (X, Δ) only depends on the linear equivalence class of $K_X + \Delta$, replacing A by an $\mathbb{R}$ -linearly equivalent divisor, we may assume that $A = A_1 + A_2$, where A_1 is an ample divisor over U and the components of A_2 generate the relative Néron-Severi group of X over U .

Let $\phi: X \dashrightarrow Y$ be a birational morphism over U , whose inverse does not contract any divisors, and let $(Y, \Gamma = \phi_*\Delta)$ be a weak log canonical model over U of the pair (X, Δ) , where $\Delta \in \mathcal{C}$. Since $\mathcal{C}$ is compact, this result is local about Δ . Thus by (3.6.3) we may assume that $K_X + \Delta$ is kawamata log terminal.

Let $p: Z \longrightarrow X$ be a log resolution of (X, Δ) which resolves the indeterminacy locus of ϕ . Let Φ be the divisor whose existence is guaranteed by (6.2). As we are assuming Theorem A_n , there is a log terminal model $\xi: Z \dashrightarrow W$ of $K_Z + \Phi$ over Y . In particular there is a morphism $f: W \longrightarrow Y$, a birational map $\psi: X \dashrightarrow W$, and $K_W + \Psi$ is nef over Y , where $\Psi = \xi_*\Phi$. By (6.2) $K_W + \Psi$ is a log terminal model of $K_X + \Delta$ over U .

Pick H an ample divisor over U on Y . As the components of ψ_*A_2 generate the relative Néron-Severi group of W over U , there is a divisor B on W with support equal to ψ_*A_2 such that B and f^*H are numerically equivalent. Then B and $B - (K_W + \Psi)$ are nef over Y , and so by the base point free theorem, cf. (3.4.1), B is the pullback of an ample divisor on Y .

Replacing H by f_*B , we may assume that $B = f^*H$, and we may also assume that

$$K_Y + \Gamma' = K_Y + \Gamma + H,$$

is kawamata log terminal and ample over U . Let

$$\Delta' = \Delta + H',$$

where H' is the strict transform of B . Possibly rescaling H we may assume that $K_X + \Delta'$ is kawamata log terminal and ψ -negative. Note that then ϕ is $(K_X + \Delta')$ -non-negative and $A_1 \leq \Delta'$. Therefore replacing A by A_1 , (Y, Γ) with (Y, Γ') and (X, Δ) by (X, Δ') , we may assume that (Y, Γ) is the log canonical model of (X, Δ) , and the result follows by (6.1). $\square$

In terms of induction, we will need a version of (6.1) locally around the locus where $K_X + \Delta$ is not kawamata log terminal. To this end we need a version of (4.2) for a convex set of divisors:

Proposition 6.4. *Assume Theorem B_{n-1} and Theorem A_n . Let $(X, \Delta_0 = S + A + B_0 \in \mathcal{P}_A)$ be a log smooth projective purely log terminal pair of dimension n . Let V_0 be the vector space of Weil divisors on X generated by the components of B_0 . Fix a general ample divisor H such that $K_X + \Delta_0 + H$ is ample, and let V be the translate by $S + A$, of vector space spanned by V_0 and H . Given any polytope F in V , the cone $\mathcal{C}(F)$ over F (with vertex Δ_0) is the polytope spanned by F and Δ_0 .*

Pick a constant $\alpha > 0$ such that for all $\|E\| < \alpha$, $E \in V_0$, $E + H$ is sufficiently ample, and let

$$F = \{ \Delta_0 + E + H \in V \mid \|E\| \leq \alpha, E \in V_0 \} \subseteq \mathcal{N}_A \quad \text{and} \quad \mathcal{C}_0 = \mathcal{C}(F).$$

Assume that $S \not\subset N_\sigma(K_X + \Delta_0)$.

If $K_X + \Delta_0$ does not have a log terminal model then there is a countable collection of polytopes $\mathcal{P}_i$ and birational maps $\phi_i: X \dashrightarrow Y_i$ such that

(1)

$$\mathcal{P}_i^0 \cap \mathcal{P}_j^0 = \emptyset \quad \text{for } i \neq j,$$

(2) $\mathcal{P}_i \subset \mathcal{P}_{Y_i}$,

(3) for any $\Delta \in F$, the $(K_X + \Delta_0)$ -MMP with scaling of $E + H$ is given by

$$X \dashrightarrow Y_{i_1} \dashrightarrow Y_{i_2} \dashrightarrow Y_{i_3} \cdots,$$

for appropriate indices i_j ,

(4) for all $\epsilon > 0$, the set

$$\{i \in I \mid \exists \Delta = \Delta_0 + t(E + H) \in \mathcal{P}_i, t > \epsilon\},$$

is finite, and

(5) if $\mathcal{C}_i$ denotes the cone over $\mathcal{P}_i$, then

$$\mathcal{C}_i - \{\Delta_0\} = \bigcup \{\mathcal{P}_j \mid \mathcal{P}_j \subset \mathcal{C}_i\}.$$

Proof. By assumption the set

$$\mathcal{P}_0 = \mathcal{C}_0 \cap \mathcal{N}_A,$$

does not contain Δ_0 . Moreover $\mathcal{P}_0$ is a polytope. Let $\phi_0: X \dashrightarrow Y_0 = X$ be the identity map. Note that $\mathcal{C}_0$ is indeed the cone over $\mathcal{P}_0$.

Suppose that we have defined $\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_k$, satisfying (1), (2) and a modified version of (3), where we stop the MMP in (3) when Δ lies on the boundary of $\cup \mathcal{P}_i$. Let $F_{i,j}$ be the faces of $\mathcal{P}_i$, whose associated cones are of maximal dimension. For each such face, let $F = F_{i,j}$ and let $\mathcal{C}_{k+1}$ be the corresponding cone. If $\mathcal{P}_i$ is of maximal dimension, we discard F whenever $\mathcal{C}_{k+1} = \mathcal{C}_i$, for some $i < k$. Let R be a $(K_X + \Delta_0)$ -extremal ray of Y_i , which cuts out F . Let $Y_i \dashrightarrow Y_{k+1}$ be the corresponding step of the $(K_X + \Delta_0)$ -MMP. As we are assuming $S \not\subset N_\sigma(K_X + \Delta_0)$ then S is not contracted by ϕ .

Let $\mathcal{P}_{k+1}$ be the nef cone of Y_{k+1} intersected with $\mathcal{C}_{k+1}$. As before $\mathcal{P}_{k+1}$ is a polytope that, by assumption, does not contain Δ_0 .

Thus, by induction, we may assume that we have constructed a countable set of polytopes $\mathcal{P}_i$ and rational maps $\phi_i: X \dashrightarrow Y_i$ satisfying (1) (2) and the modified version of (3) described above. Note that property (4) then follows from (6.3), and (3) and (5) follow from (4), since each $\mathcal{C}_i$ contains infinitely many $\mathcal{P}_j$. $\square$

Lemma 6.5. Assume Theorem B_{n-1} and Theorem A_n . Let $(X, \Delta_0 = S + A + B_0)$ be a purely log terminal pair, where X is projective of

dimension n , A is ample, $\lfloor \Delta_0 \rfloor = S$ and $B_0 \geq 0$. Suppose that S is not a component of $N_\sigma(K_X + \Delta)$.

Then we may find an ample divisor H , a positive constant α , and a log pair (W, R) , such that if V is the translate by $S + A$ of the vector space of Weil divisors on X generated by H and the components of B_0 and $B \in V_0$ (the span of the components of B_0) is such that

$$\|B - B_0\| < \alpha t,$$

for some $t \in (0, 1]$, then we may find a log terminal model $\phi: X \dashrightarrow Y$ of $(X, \Delta = S + A + B + tH)$ which does not contract S and such that the pairs (W, R) and $(Y, T = \phi_* S)$ have isomorphic neighbourhoods of R and T .

Proof. Suppose not. Passing to a log resolution, we may assume that (X, Δ_0) is log smooth. Using the notation established in (6.4) and possibly relabelling, by assumption there is no $\mathcal{C}_k$ such that for any two elements Δ_1 and Δ_2 in $\mathcal{C}_k$, the corresponding models have isomorphic neighbourhoods of T . Hence we may find a sequence of polytopes $\mathcal{P}_i$, such that the corresponding cones are nested $\mathcal{C}_i \subset \mathcal{C}_{i-1}$, and moreover the corresponding Y_i are not eventually isomorphic in a neighbourhood of S . By compactness of F , we may find $\Delta \in F$ such that $(\Delta_0, \Delta] \cap \mathcal{C}_i \neq \emptyset$ for every i . By (3), the corresponding $(K_X + \Delta_0)$ -MMP with scaling of $E + H = \Delta - \Delta_0$ is not eventually an isomorphism in a neighbourhood of S and this contradicts (4.2). $\square$

7. NON-VANISHING

We follow the general lines of the proof of the non-vanishing theorem, see for example Chapter 3, §5 of [26]. In particular there are two cases:

Lemma 7.1. *Assume Theorem A_n . Let (X, Δ) be a projective, log smooth, kawamata log terminal pair of dimension n , such that $K_X + \Delta$ is pseudo-effective and $\Delta - A \geq 0$ for an ample $\mathbb{Q}$ -divisor A . Suppose that for every positive integer k such that kA is integral,*

$$h^0(X, \mathcal{O}_X(\lfloor mk(K_X + \Delta) \rfloor + kA)),$$

is a bounded function of m .

Then there is an $\mathbb{R}$ -divisor D such that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$.

Proof. By (3.3.2) it follows that $K_X + \Delta$ is numerically equivalent to $N_\sigma(K_X + \Delta)$. Since $N_\sigma(K_X + \Delta) - (K_X + \Delta)$ is numerically trivial and ampleness is a numerical condition, it follows that

$$A' = A + N_\sigma(K_X + \Delta) - (K_X + \Delta),$$

is ample and numerically equivalent to A . Thus

$$K_X + \Delta' = K_X + A' + (\Delta - A),$$

is numerically equivalent to $K_X + \Delta$, and

$$K_X + \Delta' \sim_{\mathbb{R}} N_{\sigma}(K_X + \Delta) \geq 0.$$

Thus by Theorem A_n $K_X + \Delta'$ has a log terminal model $\phi: X \dashrightarrow Y$, which is then automatically a log terminal model for $K_X + \Delta$. Replacing (X, Δ) by (Y, Γ) we may therefore assume that $K_X + \Delta$ is nef and the result follows by the base point free theorem, cf. (3.4.1). $\square$

Lemma 7.2. *Let (X, Δ) be a projective, log smooth, kawamata log terminal pair such that $\Delta = A + B$, where A is an ample $\mathbb{Q}$ -divisor and B is effective. Suppose that there is a positive integer k such that kA is integral and*

$$h^0(X, \mathcal{O}_X(\lfloor mk(K_X + \Delta) \rfloor + kA)),$$

is an unbounded function of m .

Then we may find a projective, log smooth, purely log terminal pair (Y, Γ) and an ample $\mathbb{Q}$ -divisor C on Y , where

- Y is birational to X ,
- $\Gamma - C \geq 0$, and
- $T = \lfloor \Gamma \rfloor$ is an irreducible divisor, which is not a component of $N_{\sigma}(K_Y + \Gamma)$.

Moreover the pair (Y, Γ) has the property that $K_X + \Delta \sim_{\mathbb{R}} D \geq 0$ for some $\mathbb{R}$ -divisor D iff $K_Y + \Gamma \sim_{\mathbb{R}} G \geq 0$ for some $\mathbb{R}$ -divisor G .

Proof. Pick m large enough so that

$$h^0(X, \mathcal{O}_X(\lfloor mk(K_X + \Delta) \rfloor + kA)) > \frac{(kn)^n}{n!}.$$

By standard arguments, given any point $x \in X$, we may find an effective divisor in the linear system

$$|\lfloor mk(K_X + \Delta) \rfloor + kA|,$$

of multiplicity greater than kn at x . In particular, we may find an effective $\mathbb{R}$ -divisor

$$H \sim_{\mathbb{R}} m(K_X + \Delta) + A,$$

of multiplicity greater than n at x . Given $t \in [0, m]$, consider

$$\begin{aligned} (t+1)(K_X + \Delta) &= K_X + \frac{m-t}{m}A + B + t(K_X + \Delta + \frac{1}{m}A) \\ &\sim_{\mathbb{R}} K_X + \frac{m-t}{m}A + B + \frac{t}{m}H \\ &= K_X + \Delta_t. \end{aligned}$$

Fix $0 < \epsilon \ll 1$, let $A' = \epsilon/mA$ and $u = m - \epsilon$. We have:

- (1) $K_X + \Delta_0$ is kawamata log terminal,
- (2) $\Delta_t \geq A'$, for any $t \in [0, u]$ and
- (3) the locus of log canonical singularities of (X, Δ_u) is not contained in $N_\sigma(K_X + \Delta_u)$.

Let $\pi: Y \longrightarrow X$ be a common log resolution of (X, Δ_t) . We may write

$$K_Y + \Psi_t = \pi^*(K_X + \Delta_t) + E_t,$$

where E_t and Ψ_t are effective, with no common components, $\pi_*\Psi_t = \Delta_t$ and E_t is exceptional. Pick an effective exceptional divisor F and a positive integer l such that $l(\pi^*A' - F)$ is very ample and let lC be a very general element of the linear system $|l(\pi^*A' - F)|$. For any $t \in [0, u]$, let

$$\Gamma_t = \Psi_t - \pi^*A' + C + F \sim_{\mathbb{R}} \Psi_t.$$

After cancelling common components of Γ_t and $N_\sigma(K_Y + \Gamma_t)$, properties (1-3) above become

- (1) $K_Y + \Gamma_0$ is kawamata log terminal,
- (2) $\Gamma_t \geq C$, for any $t \in [0, u]$ and
- (3) the locus of log canonical singularities of (Y, Γ_u) is not contained in $N_\sigma(K_Y + \Gamma_u)$.

Moreover

- (4) (Y, Γ_t) is log smooth, for any $t \in [0, u]$.

Let

$$s = \sup\{t \in [0, u] \mid K_Y + \Gamma_t \text{ is log canonical}\}.$$

Thus, setting $\Gamma = \Gamma_s$, we may write

$$\Gamma = T + C + B',$$

where $\lfloor \Gamma \rfloor = T$, C is ample and B' is effective. Possibly perturbing Γ , we may assume that T is irreducible, so that $K_Y + \Gamma$ is purely log terminal and we may assume that C is $\mathbb{Q}$ -Cartier. $\square$

We will need the following consequence of Kawamata-Viehweg vanishing:

Lemma 7.3. *Let $(X, \Delta = S + A + B)$ be a $\mathbb{Q}$ -factorial projective purely log terminal pair and let m be a positive integer. Suppose that*

- (1) $S = \lfloor \Delta \rfloor$ is irreducible,
- (2) $m(K_X + \Delta)$ is integral,
- (3) $m(K_X + \Delta)$ is Cartier in a neighbourhood of S ,
- (4) $h^0(S, \mathcal{O}_S(m(K_X + \Delta))) > 0$,
- (5) $K_X + G + B$ is klt, where $G \geq 0$,

- (6) $A \sim_{\mathbb{Q}} (m-1)tH + G$ for some t ,
(7) $K_X + \Delta + tH$ is big and nef.

Then $h^0(X, \mathcal{O}_X(m(K_X + \Delta))) > 0$.

Proof. Considering the long exact sequence associated to the restriction exact sequence,

$$0 \longrightarrow \mathcal{O}_X(m(K_X + \Delta) - S) \longrightarrow \mathcal{O}_X(m(K_X + \Delta)) \longrightarrow \mathcal{O}_S(m(K_X + \Delta)) \longrightarrow 0,$$

it suffices to observe that

$$H^1(X, \mathcal{O}_X(m(K_X + \Delta) - S)) = 0,$$

by Kawamata-Viehweg vanishing, since

$$\begin{aligned} m(K_X + \Delta) - S &= (m-1)(K_X + \Delta) + K_X + A + B \\ &= K_X + G + B + (m-1)(K_X + \Delta + tH) \end{aligned}$$

and $K_X + \Delta + tH$ is big and nef. $\square$

Lemma 7.4. *Theorem A_n and Theorem B_n imply Theorem C_n .*

Proof. By (3.2.1), it suffices to prove this result for the generic fibre of U . Thus we may assume that U is a point, so that X is a projective variety.

By (3.5.5) we may assume that (X, Δ) is log smooth. By (3.6.3) we may assume that $\Delta = A + B$, where A is ample and $B \geq 0$. By (7.1) and (7.2), we may therefore assume that $\Delta = S + A + B$, where A is an ample $\mathbb{Q}$ -divisor, B is effective and $\lfloor \Delta \rfloor = S$ is irreducible and not a component of $N_\sigma(K_X + \Delta)$.

Let H be the ample divisor on X and $\alpha > 0$ be the constant whose existence is guaranteed by (6.5). Possibly replacing A by an $\mathbb{R}$ -linearly equivalent divisor, we may assume that there is a positive constant ϵ such that $A - \epsilon H \geq 0$. Let V be the affine subspace of the vector space of Weil divisors on X , generated by divisors

$$\Psi = S + A + B' + tH,$$

where $t \in \mathbb{R}$ and the support of B' is contained in the support of B .

Let $\phi: X \dashrightarrow Y$ be the log terminal model of $K_X + \Psi$, let T be the strict transform of S , for $\|B' - B\| < \alpha t$, let $\Gamma = \phi_* \Psi$ and define Θ by adjunction

$$(K_Y + \Gamma)|_T = K_T + \Theta.$$

By linearity we may formally extend the assignment $\Psi \longrightarrow \Theta$ to a rational affine linear map

$$L: V \longrightarrow W,$$

to the whole of V , where W is an appropriate finite dimensional affine space of Weil divisors on T . In particular, $L(\Delta)$ is big and by (6.5) it follows that $K_T + L(\Delta)$ is nef. Now the nef cone of T is locally polyhedral about $L(\Delta)$, by (1) of (3.6.3) and (3.9.1).

We may find a positive integer k such that if $r(K_X + \Psi)$ is integral then $rk(K_X + \Psi)$ is Cartier in a neighbourhood of T . By Kollár's effective base point free theorem, [24], we may find a positive integer M' such that if D is a nef Cartier divisor on T such that $aD - (K_T + \Theta)$ is nef and big for some $a > 0$, where $K_T + \Theta$ is kawamata log terminal, then $M'D$ is base point free, where M' only depends on a . Set $M = kM'$. In particular, if $rk(K_T + \Theta)$ is Cartier, nef, kawamata log terminal and $\Theta - \epsilon(\phi_*H)|_T \geq 0$, then

$$rk(K_T + \Theta) = K_T + (\Theta - \epsilon(\phi_*H)|_T) + C,$$

where $C = (rk - 1)(K_T + \Theta) + \epsilon(\phi_*H)|_T$ and $0 < \epsilon \ll 1$. Now if C is nef then $Mr(K_T + \Theta)$ is base point free.

By (3.9.1), there exists a rational polytope $\mathcal{C}$ in V such that $\Delta \in \mathcal{C}$ and $L(\mathcal{C}) = \mathcal{N}_{A'}$. Thus, by (3.6.4), there are real numbers $r_i > 0$ with $\sum r_i = 1$, a positive integer $p > 0$ and $\mathbb{Q}$ -divisors $\Delta_i \in \mathcal{C}$ such that

$$p(K_X + \Delta),$$

is integral,

$$K_X + \Delta = \sum r_i(K_X + \Delta_i),$$

and

$$\|\Delta - \Delta_i\| \leq \frac{\alpha}{m},$$

where $m = Mp$. We may assume that $\frac{1}{m-1} < \epsilon$. Let $\Theta_i = L(\Delta_i)$.

By our choice of k , $pk(K_T + \Theta_i)$ is Cartier. By our choice of $\mathcal{C}$, $K_T + \Theta_i + t(\phi_*H)|_T$ is nef for any $t \geq 0$ sufficiently small and $\Theta_i \geq \epsilon(\phi_*H)|_T$, which is big, so that by our choice of M' ,

$$h^0(T, \mathcal{O}_T(m(K_T + \Theta_i))) > 0.$$

(7.3) implies that $h^0(Y, \mathcal{O}_Y(m(K_Y + \Gamma_i))) > 0$, where $\Gamma_i = \phi_*\Delta_i$. As ϕ is $(K_X + \Delta_i + tH)$ -negative, it is certainly $(K_X + \Delta_i)$ -negative. But then

$$h^0(X, \mathcal{O}_X(m(K_X + \Delta_i))) = h^0(Y, \mathcal{O}_Y(m(K_Y + \Gamma_i))) > 0.$$

In particular there is an $\mathbb{R}$ -divisor D such that

$$K_X + \Delta = \sum r_i(K_X + \Delta_i) \sim_{\mathbb{R}} D \geq 0. \quad \square$$

8. PROOF OF THEOREMS

Proof of Theorem A, Theorem B and Theorem C. Immediate from (5.9), (6.3) and (7.4). $\square$

Proof of (1.2). By (3.5.5) we may assume that π is projective. We first prove (1) when there is a finite Galois étale cover $V \rightarrow U$, with Galois group G , where V is a quasi-projective variety. Let X' be the fibre square, and let Δ' be the pullback of Δ to X' . Then there is natural bijection between the log terminal models of $K_X + \Delta$ and the G -equivariant log terminal models of $K_{X'} + \Delta'$. If $K_X + \Delta$ is big over U , then by (3.2.1), we may pick A ample over U and $B \geq 0$, such that $K_X + \Delta \sim_{\mathbb{R}, U} A + B$. Let

$$\Psi = \Delta + \epsilon(A + B).$$

Then $K_X + \Psi$ is kawamata log terminal for ϵ sufficiently small and by construction Ψ is big over U . On the other hand a log terminal model for $K_X + \Psi$ is automatically a log terminal model for $K_X + \Delta$. Thus we may assume that Δ is big over U , and the result follows from Theorem A and Theorem C, when U has a finite étale cover which is a quasi-projective variety.

We now prove (1) in the general case. By (3.5.5), we may assume that (X, Δ) is log smooth. Let U_0 be any open subset of U . Suppose that there is a log terminal model $\phi_0: X_0 \dashrightarrow Y_0$ for $K_{X_0} + \Delta_0$ over U_0 , where X_0 is the inverse image of U_0 and Δ_0 is the restriction of Δ to X_0 . Let U_1 be any open subset of U , which has a finite cover which is affine. Pick A_0 an ample divisor over U_0 on Y_0 . Then $K_{Y_0} + \Gamma_0 + \epsilon A_0$ is the log canonical model for $K_{X_0} + \Delta_0 + \epsilon B_0$ over U_0 for any $\epsilon > 0$, where $B_0 = \phi_0^* A_0$. In particular, Y_0 is the unique log terminal model of $K_{X_0} + \Delta_0 + \epsilon B_0$.

Let B be the closure of B_0 in X . By what we have already proved we may find $\phi_1: X_1 \dashrightarrow Y_1$ a log terminal model for $K_{X_1} + \Delta_1 + \epsilon B_1$ over U_1 , where B_1 is the restriction to U_1 of B . By finiteness, we may assume that ϕ_1 is independent of $\epsilon > 0$, for ϵ sufficiently small. On the other hand, the restrictions of Y_0 and Y_1 to the inverse image of $U_0 \cap U_1$ are isomorphic, by uniqueness of the log terminal model for $K_{X_0} + \Delta_0 + \epsilon B_0$. Let Y_2 be the algebraic space obtained by gluing Y_0 to Y_1 along the inverse images of $U_0 \cap U_1$. Let $U_2 = U_0 \cup U_1$, let X_2 be the inverse image of U_2 and let Δ_2 and B_2 be the restriction of Δ and B to X_2 . Then $Y_2 \rightarrow U_2$ is projective (so that in particular Y_2 is separated) and so Y_2 is a log terminal model for $K_{X_2} + \Delta_2 + \epsilon B_2$ over U_2 , for any $\epsilon > 0$. Taking the limit as ϵ goes to zero, Y_2 is then

a log terminal model for $K_{X_2} + \Delta_2$ over U_2 . (1) follows by Noetherian induction.

(2) follows from (1) and the base point free theorem, (3.4.1) and (3) is immediate from (2). $\square$

9. PROOF OF COROLLARIES

Proof of (1.1.1). (1.2) implies that there is a log terminal model $\phi: X \dashrightarrow Y$ for K_X , which is then automatically a minimal model. Hence (1). (2) follows from the base point free theorem and (1), and then (3) is immediate. (4) is Theorem D of [7]. $\square$

Proof of (1.1.2). Immediate by Theorem (5.2) of [10] and (3) of (1.2). $\square$

Proof of (1.1.3). It is well known that Y_1 and Y_2 are isomorphic in codimension one (cf. [14] and [23]). Let H be a divisor on Y_2 , which is ample over U . Let G be the strict transform on Y_1 . Possibly replacing G by a small multiple, we may assume that $K_{Y_1} + \Gamma_1 + G$ is kawamata log terminal. Since $Y_1 \dashrightarrow Y_2$ is an isomorphism iff it is a morphism, as Y_2 is $\mathbb{Q}$ -factorial, then $Y_1 \dashrightarrow Y_2$ is an isomorphism iff G is nef over U (by negativity of contraction), which in turn is equivalent to $K_{Y_1} + \Gamma_1 + G$ being nef over U .

Consider running the directed $(K_{Y_1} + \Gamma_1 + G)$ -MMP over U , where we only consider extremal rays on which $K_{Y_1} + \Gamma_1$ is trivial. Suppose that $K_{Y_1} + \Gamma_1 + G$ is not nef over U . As $K_{Y_1} + \Gamma_1$ is nef over U , it follows that $K_{Y_1} + \Gamma_1 + \epsilon G$ is not nef for all $\epsilon > 0$. But then we may find an extremal ray which is $(K_{Y_1} + \Gamma_1 + G)$ -negative and $(K_{Y_1} + \Gamma_1)$ -trivial. By (1.4.2) the directed MMP $(K_{Y_1} + \Gamma_1 + G)$ -MMP over U then ends when $K_{Y_1} + \Gamma_1 + G$ is nef over U . $\square$

Proof of (1.1.5). We first prove (1) and (2). By Theorem B and (3.9.2), and since log canonical models are unique, it suffices to prove that if $\Delta \in \mathcal{P}_A$ then $K_X + \Delta$ has a weak log canonical model.

By (2) of (3.6.3) we may assume that $K_X + \Delta$ is kawamata log terminal, and we may apply (1.2). Thus we may assume that $K_X + \Delta$ is nef over U and the result follows the base point free theorem, cf. (3.4.1).

We now prove (3). Let Δ_k , $k = 1$ and 2 , belong to the interior of the same face of $\mathcal{P}_i$. It suffices to prove that the ample model of $K_X + \Delta_k$ is independent of k . Let $\Gamma_k = \phi_{i*}\Delta_k$. Replacing X by Y we may assume that $K_X + \Delta_k$ is nef.

Then $K_X + \Delta_k$ are zero on precisely the same curves. On the other hand, the ample model is determined by exactly these curves. $\square$

Proof of (1.1.7). Immediate consequence of (1.1.5). $\square$

Proof of (1.1.9). Let $\psi_j: X \dashrightarrow Z_j$, $1 \leq j \leq q$ be the finitely many ample models whose existence is guaranteed by (1.1.5). Let $\mathcal{C}$ be the polytope spanned by $\Delta_1, \Delta_2, \dots, \Delta_k$ and let $\mathcal{C}_j$ be subpolytope given by the closure of those divisors with ample model ψ_j .

Replacing $\Delta_1, \Delta_2, \dots, \Delta_k$ by the extreme points of $\mathcal{C}_j$, we may assume that $\mathcal{C} = \mathcal{C}_j$ and so we may drop the index j . Pick m_i such that $D_i = m_i(K_X + \Delta_i)$ is integral. Passing to a truncation it suffices to prove that

$$\mathcal{R}(\pi, D^\bullet) = \bigoplus_{m \in \mathbb{N}^k} \pi_* \mathcal{O}_X(\sum m_i D_i),$$

is a finitely generated $\mathcal{O}_U$ -module. Passing to a truncation, we may assume that

$$D_i = \psi^* H_i,$$

where H_i is semiample on Z . In this case

$$\mathcal{R}(\pi, D^\bullet) = \mathcal{R}(\pi', H^\bullet).$$

But the latter is finitely generated since it is a semiample vector bundle. $\square$

Proof of (1.2.2). We follow the argument given in [17]. Fix positive integers r and n and a positive real number d . Consider the family of pairs (X, Δ) such that

- (i) X has dimension n ,
- (ii) $K_X + \Delta$ is kawamata log terminal,
- (iii) $K_X + \Delta$ is ample,
- (iv) $r(K_X + \Delta)$ is Cartier, and
- (v) the degree $(K_X + \Delta)^n < d$.

By the effective base point theorem of Kollár, see [24], there is a positive integer M such that if (X, Δ) is any pair satisfying (i-iv) then $M(K_X + \Delta)$ is very ample. Thus for the family of all pairs (X, Δ) satisfying (i-v) the pairs $(X, \mathcal{O}_X(M(K_X + \Delta)))$ belong to a bounded family. It is then easy to see that the pairs (X, Δ) themselves also belong to a bounded family.

Thus we may find a pair (Y, Γ) and a projective family $\pi: Y \rightarrow U$, over a quasi-projective base, such that every pair (X, Δ) is isomorphic to a fibre of this family, and any fibre over an open subset is a pair satisfying (i-v). Pick any compactification of this family such that U is projective. Note that any pair (X, Δ) satisfying (i-iv) is its own log canonical model. Since the log canonical model is unique, passing to

a semi-stable reduction and taking the log canonical model we may assume that (Y, Γ) is terminal.

Thus the result follows from (3.4.2) and the main result of [27]. $\square$

Proof of (1.2.3). Immediate consequence of Theorem A, Theorem B and Theorem C. $\square$

Proof of (1.3.1). Immediate from (1.4.2) and the main result of [16]. $\square$

Proof of (1.3.2). Let A be any π -ample divisor, and let c be the infimum such that $K_X + \Delta + cA$ is π -pseudo-effective. Then $c > 0$ as $K_X + \Delta$ is not π -pseudo-effective and as c is an infimum, $K_X + \Delta + cA$ is not π -big.

Consider running the $(K_X + \Delta + cA)$ -MMP over U with scaling of A . At the end we arrive at a model $\phi: X \dashrightarrow Y$, such that $K_Y + \Gamma + cA'$ is nef, where A' is the image of A . Since every step of this MMP is automatically $(K_X + \Delta)$ -negative, replacing (X, Δ) by (Y, Γ) , we may assume that $K_X + \Delta + cA$ is nef (of course in this case A is no longer ample, it is only big).

Applying the base point free theorem to $K_X + \Delta + cA$ we get a contraction morphism $f: X \rightarrow W$ over U , such that $K_X + \Delta + cA$ is the pullback of a relatively ample divisor. Replacing U by W we may therefore assume that $K_X + \Delta + cA$ is the pullback of an ample divisor from U . Consider continuing to run the $(K_X + \Delta + cA)$ -MMP with scaling of A . Clearly we never change the value of c , and this program only terminates when we arrive at a Mori fibre space for $K_Y + \Gamma$. $\square$

Proof of (1.4.1). The flip of π is precisely the log canonical model, so that this result follows from Theorem A. $\square$

Proof of (1.4.2). It suffices to prove that we cannot find an infinite sequence of flips for the directed MMP, and this follows from (4.1). $\square$

Proof of (1.4.3). Perturbing Δ by an ample divisor which contains the centre of every element of $\mathcal{E}$ of log discrepancy one, we may assume that $\mathcal{E}$ contains no exceptional divisors of log discrepancy one. Replacing Δ by $(1 - \epsilon)\Delta + \epsilon\Delta_0$ we may assume that $K_X + \Delta$ is kawamata log terminal. We may write

$$K_Y + \Psi = f^*(K_X + \Delta) + E,$$

where Ψ and E are effective divisors, with no common components, $f_*\Psi = \Delta$ and E is exceptional. Let F be the reduced sum of all the exceptional divisors which are not components of E nor elements of $\mathcal{E}$.

Pick $\epsilon > 0$ such that $K_Y + \Gamma = K_Y + \Psi + \epsilon F$ is kawamata log terminal. As f is birational Γ is big over X and so by (1.2) we may find a log terminal model $\phi: Y \rightarrow W$ over X for (Y, Γ) .

Replacing Y by W we may assume that $K_Y + \Gamma$ is nef over U . By negativity of contraction it follows that $E + \epsilon F$ is zero, so that

$$K_Y + \Psi = f^*(K_X + \Delta).$$

But then we must have contracted every exceptional divisor which does not belong to $\mathcal{E}$. $\square$

Proof of (1.4.4). Since (X, Δ) is purely log terminal near the image of S if and only if (S, Θ) is kawamata log terminal by (17.6) of [25], and since (X, Δ) is log canonical near the image of S if and only if (S, Θ) is log canonical (cf. [18]), we may assume that (X, Δ) is purely log terminal near the image of S and (S, Θ) is kawamata log terminal. At this point we simply follow the proof of (17.12) of [25], substituting (1.4.3) for (17.10) of [25]. $\square$

Proof of (1.4.5). Let $f: Y \rightarrow X$ be a log resolution of (X, Δ) , so that the composition $\psi: Y \rightarrow U$ of f and π is projective. We may write

$$K_Y + \Gamma = f^*(K_X + \Delta) + E,$$

where Γ and E are effective with no common components and E is f -exceptional.

Since $K_X + \Delta$ is π -nef it is π -pseudo-effective and so $K_Y + \Gamma$ is ψ -pseudo-effective. Pick a ψ -ample divisor A . Then for every positive real number $t > 0$, $K_X + \Delta + tA$ is π -big. Let $\psi_t: X_t \rightarrow U$ be the corresponding log canonical model. As X contains no rational curves which are contracted by π , the induced rational map $f_t: X_t \dashrightarrow X$ is a morphism. Since f_t is birational, Theorem B implies that there are only finitely many X_t up to isomorphism. Let X' be the limit as t goes to zero. Replacing Y by X' we may assume that $K_Y + \Gamma$ is f -nef, so that E is empty, and f is small. As X is $\mathbb{Q}$ -factorial it follows that f is an isomorphism. But then π is a log terminal model. $\square$

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